Renormalization

1. Renormalization in ϕ^3-theory

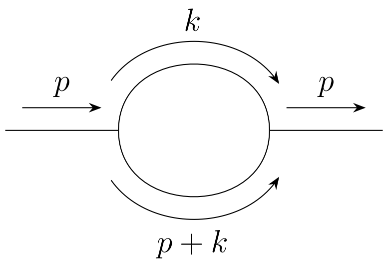
One loop of self energy

Simply speaking, renormalization is to solve the problem of divergent when we try to solve the loop graphs by subtracting the divergent part in the expression. In the following, we will use ϕ^3-theory as an example to discuss the renormalization.

Firstly, the Lagrangian of ϕ^3-theory is:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3. \tag{2.1}$$

As the simplest case, the 1-loop correlation for the propagator is (notice that in this graph ,there are **all inner lines**, i.e., propagators):



where the momentum k is **off-shelled**, which means k doesn't follow the relation $k^2+E_k^2 = m^2$, so technically k could be very large. But we don't need to worry about it, because k only appears in the loop, which means that it must follows the 4-momentum conservation. Thus we need to sum over all possible k values, which finally will contribute a integral measurement when we calculate the loop part. Thus, generally, we can write the whole graph as

$$\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}\left[-\mathrm{i}\Sigma_1(p^2)\right]\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}, \tag{2.2}$$

where $-\mathrm{i}\Sigma_1(p^2)$ represent the loop part, i.e.,

$$-\mathrm{i}\Sigma_1(p^2) = \frac{g^2}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(p+k)^2 - m^2 + \mathrm{i}\epsilon}, \tag{2.3}$$

where we will note the integral measurement $\frac{\mathrm{d}^4k}{(2\pi)^4} \to \mathrm{d}^4k$ for simple. Next, we will focus on this integral as an example to talk about how to solve this type of integral.

1.1 Wick rotation

The integral in (2.3) over the loop energy can be also written as

$$I(p,\vec{k}) = \int_{-\infty}^{+\infty}\mathrm{d}k^0\,\frac{\mathrm{i}}{(k^0)^2 - E_k^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(p^0+k^0)^2 - E_{p+k}^2 + \mathrm{i}\epsilon}, \tag{2.4}$$

where

$$E_k = \sqrt{\vec{k}^{\,2} + m^2}, \qquad E_{p+k} = \sqrt{(\vec{p} + \vec{k})^2 + m^2}. \tag{2.5}$$

The variable k^0 is now regarded as a complex variable. The four poles are

$$k^0 = +E_k - \mathrm{i}\epsilon, \qquad k^0 = -E_k + \mathrm{i}\epsilon, \tag{2.6}$$

$$k^0 = -p^0 + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon. \tag{2.7}$$

The small $\mathrm{i}\epsilon$ is important: it tells us which poles sit slightly above or slightly below the real axis. Wick rotation means changing the integration path in the complex k^0 plane. Originally the integral is along the real axis,

$$C_{\mathrm{M}}: \quad k^0 \in (-\infty, +\infty).$$

In Euclidean variables we use

$$k^0 = \mathrm{i}\omega, \qquad \mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega,$$

so the new path is the imaginary axis,

$$C_{\mathrm{E}}: \quad k^0 = \mathrm{i}\omega, \quad \omega \in (-\infty, +\infty).$$

In the figures below, the blue path is the real-axis Minkowski contour C_{M}, the green path is the imaginary-axis Euclidean contour C_{E}, and the crosses are poles.

There is one orientation point which is very easy to mix up. The Euclidean contour itself is oriented upward, because

$$\omega : -\infty \to +\infty \quad \Longrightarrow \quad k^0 = \mathrm{i}\omega : -\mathrm{i}\infty \to +\mathrm{i}\infty.$$

But when we prove the Wick rotation by closing the contour, the closed contour is

$$C_{\mathrm{closed}} = C_{\mathrm{M}} + A_+ - C_{\mathrm{E}} + A_-,$$

where A_+ runs from $+R$ to $+\mathrm{i}R$ in the first quadrant, while A_- runs from $-\mathrm{i}R$ back to $-R$ in the third quadrant. Thus the imaginary-axis part of the closed contour is the downward path $-C_{\mathrm{E}}$. After moving it to the other side of the equation, the final Euclidean integral is again along C_{E}, upward. The orange dashed arrows in the figures show these auxiliary closing arcs, not the direction of the final integral along C_{E}.

The phrase "change the contour" means the following simple thing: we slide the integration line from one path to another in the complex k^0 plane. If the sliding line does not hit any pole, the integral is unchanged, apart from the Jacobian $\mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega$. If the line hits a pole, then we must bend the path around that pole, and the difference is a residue term. Symbolically,

$$\int_{C_{\mathrm{old}}} f(k^0)\,\mathrm{d}k^0 - \int_{C_{\mathrm{new}}} f(k^0)\,\mathrm{d}k^0 = 2\pi\mathrm{i} \sum_{\text{poles swept by the contour}} \operatorname{Res} f, \tag{2.8}$$

up to the sign determined by the orientation of the contour.

Case 1: p^0 is imaginary.

Take, for example, $p^0 = -\mathrm{i}P$ with $P > 0$. Then $-p^0 = +\mathrm{i}P$, so the two poles from the second propagator are shifted upward:

$$k^0 = \mathrm{i}P + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = \mathrm{i}P - E_{p+k} + \mathrm{i}\epsilon.$$

This is the Euclidean starting point. The current integration path is C_{E}, not C_{M}.

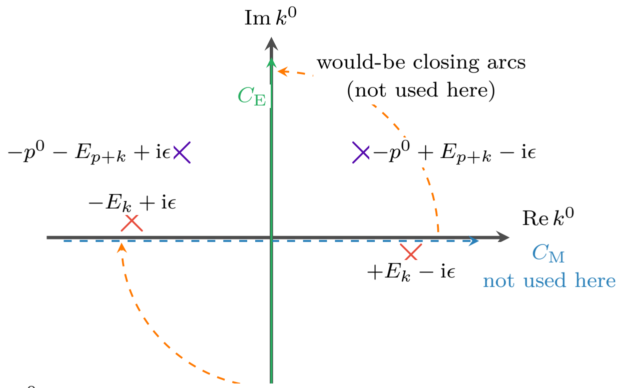


Fig. 1: p^0 imaginary. The first-quadrant pole is harmless, because the contour currently being used is only the green line C_{E}.

The point which is easy to miss is this: the pole in the first quadrant does not automatically give a residue. A residue appears only when the contour is actually moved through that pole, or when the pole moves through the contour. In this first figure we are not rotating C_{E} into C_{M} while keeping p^0 imaginary. We are simply defining the Euclidean integral on the green line.

Case 2: p^0 is real and $|p^0| < m$.

Now keep the green contour C_{E} fixed, and analytically continue p^0 from imaginary value back to a real value. The poles move because their positions contain p^0. If $|p^0| < m$, then

$$E_{p+k} \geq m > |p^0|,$$

so

$$-p^0 + E_{p+k} > 0, \qquad -p^0 - E_{p+k} < 0.$$

Thus the second pair of poles ends up in the same pattern as the first pair: one pole is slightly below the real axis on the right, and one pole is slightly above the real axis on the left.

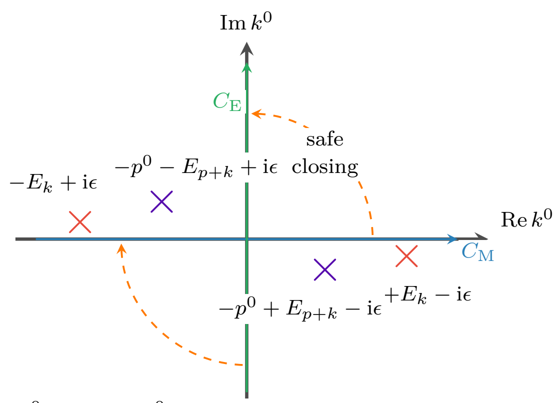


Fig. 2: p^0 real and $|p^0| < m$. No pole lies in the region swept when the blue contour is rotated into the green contour.

Therefore, for $|p^0| < m$, the real-axis integral and the imaginary-axis integral are related by an ordinary Wick rotation. On the green contour,

$$k^0 = \mathrm{i}\omega, \qquad k^2 = (k^0)^2 - \vec{k}^{\,2} = -(\omega^2 + \vec{k}^{\,2}) < 0,$$

and similarly $(p+k)^2 < 0$ in the large Euclidean region. This is why power counting becomes Euclidean.

What does "a pole crosses a contour" mean?

A pole is not just a fixed dot: its position depends on the external energy p^0. For example, if $p^0 = -P$ with $P > 0$, then one pole is

$$k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon = P - E_{p+k} + \mathrm{i}\epsilon. \tag{2.9}$$

As P increases, this pole moves horizontally. If $P < E_{p+k}$, it is on the left side of the imaginary axis. If $P = E_{p+k}$, it sits on the imaginary-axis contour. If $P > E_{p+k}$, it has moved to the right side of the imaginary axis. This is what we mean by saying that the pole has crossed C_{E}.

In formula language, crossing the imaginary axis means

$$\mathrm{Re}\, k^0_{\text{pole}} \quad \text{changes sign.}$$

Crossing the real axis would similarly mean

$$\mathrm{Im}\, k^0_{\text{pole}} \quad \text{changes sign.}$$

But only crossing the contour that we are actually using produces a residue.

Case 3: p^0 is real and $|p^0| > m$.

Take the negative-energy example $p^0 = -P$ with $P > m$, which is the case drawn in the book. The pole

$$k^0 = P - E_{p+k} + \mathrm{i}\epsilon$$

crosses the green contour whenever

$$P > E_{p+k} \quad \Longleftrightarrow \quad (p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

When this happens, the contour cannot pass straight through the pole. It must detour around it, and this detour gives an extra residue, usually called the pole term.

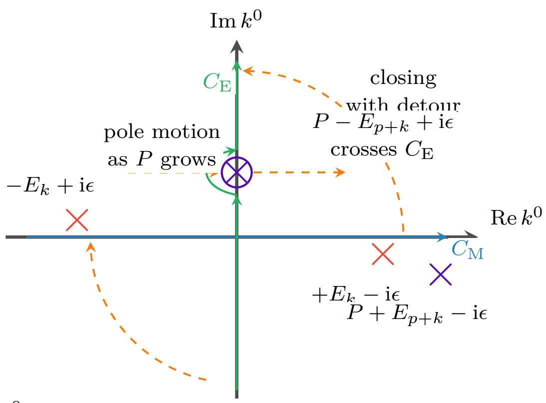


Fig. 3: $p^0 = -P$ and $P > m$. A pole crosses the green contour, so the Wick-rotated answer contains an extra residue term.

However, this pole term can occur only when

$$(p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

For very large $|\vec{k}|$, the right-hand side is very large, so the inequality cannot hold. Therefore the pole term comes only from a finite region of $\vec{k}$, not from the UV region $|\vec{k}| \to \infty$. The UV divergence is still captured by the Euclidean large-momentum integral along C_{E}.

So the divergence in Eq.(2.3) is called the UV divergence, which can be seen by firstly taking the polar coordinate: $\mathrm{d}^4 k_E = k_E^3 \mathrm{d}k_E \mathrm{d}\Omega$, thus in large k

$$\Sigma_1 \sim \int \mathrm{d}^d k \frac{1}{k^4} \sim \int \mathrm{d}k \mathrm{d}\Omega \frac{1}{k} \sim \int \frac{\mathrm{d}k}{k} \sim \ln \Lambda \to \infty. \tag{2.10}$$

1.2 On-shell renormalization

Renormalization begins by separating a loop integral into two parts: a finite, momentum-dependent part and a divergent part that is local. For the one-loop self-energy in ϕ^3 theory,

$$\Sigma_1(p^2, m^2) = \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4 k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.11}$$

The integral is logarithmically divergent in the ultraviolet. A convenient way to display the divergent piece is to add and subtract a simple reference integral. We introduce an arbitrary finite mass scale μ and write

$$\begin{aligned}\Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4} \bigg[\int \mathrm{d}^4 k \bigg\{ &\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \bigg\} \\ &+ \int \mathrm{d}^4 k \, \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \bigg].\end{aligned} \tag{2.12}$$

This is just adding zero, so no physics has been changed. The reason for this particular subtraction is that the first integral now falls faster at large loop momentum:

$$\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} = O\left(\frac{1}{k^6}\right). \tag{2.13}$$

In four dimensions, $\mathrm{d}^4 k \sim k^3 \mathrm{d}k$, and therefore

$$\int^{\infty} \mathrm{d}^4 k \, O\left(\frac{1}{k^6}\right) \sim \int^{\infty} \mathrm{d}k \, \frac{1}{k^3}, \tag{2.14}$$

which is ultraviolet finite. Thus the logarithmic divergence has been isolated in the second integral. Since that second integral has no dependence on the external momentum p, its divergent part can be canceled by a local mass counterterm.

To see the divergence explicitly, regulate the theory on a spacetime lattice with lattice spacing a. The lattice spacing acts as a UV cutoff of order $1/a$. If $S_F(k; m, a)$ denotes the regulated free propagator, then

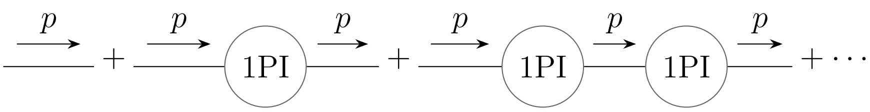
$$\begin{aligned}\Sigma_1 &= \frac{\mathrm{i}g^2}{32\pi^4} \left[\int \mathrm{d}^4 k \, \{ S_F(k; m, a) S_F(p+k; m, a) - S_F^2(k; \mu, a) \} + \int \mathrm{d}^4 k \, S_F^2(k; \mu, a) \right] \\ &\equiv \Sigma_{1,\mathrm{fin}}(p, m, \mu, a) + \Sigma_{1,\mathrm{div}}(\mu, a).\end{aligned} \tag{2.15}$$

The divergent reference part is

$$
\begin{aligned}
\Sigma_{1,\mathrm{div}}(\mu, a) &= \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k\, S_F^2(k;\mu,a) \\
&= -\frac{\mathrm{i}g^2}{32\pi^4} \int_{\omega^2+\vec{k}^{\,2}<1/a^2} \mathrm{d}\omega\, \mathrm{d}^3\vec{k}\, \frac{1}{(\omega^2+\vec{k}^{\,2}+\mu^2)^2} \\
&= -\frac{\mathrm{i}g^2}{16\pi^2} \int_0^{1/a} \mathrm{d}K\, \frac{K^3}{(K^2+\mu^2)^2} \\
&= -\frac{\mathrm{i}g^2}{16\pi^2} \ln\frac{1}{a} + \text{finite terms}, \qquad a \to 0.
\end{aligned} \tag{2.16}
$$

This agrees with the power-counting estimate in Eq. (2.10): the one-loop two-point graph in four-dimensional ϕ^3 theory is logarithmically divergent.

The same divergence can be understood from the full propagator. Let $-\mathrm{i}\Sigma(p^2)$ denote the sum of all one-particle-irreducible two-point insertions. Repeated insertions give

$G_2(p^2) =$ 

$$
\begin{aligned}
&= \frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \cdots \\
&= \frac{\mathrm{i}}{p^2-m^2}\left[1 + \frac{\Sigma(p^2)}{p^2-m^2} + \left(\frac{\Sigma(p^2)}{p^2-m^2}\right)^2 + \cdots\right] \\
&= \frac{\mathrm{i}}{p^2-m^2-\Sigma(p^2)}.
\end{aligned} \tag{2.17}
$$

The physical mass is defined by the pole of the full propagator:

$$
m_{\mathrm{ph}}^2 - m^2 - \Sigma(m_{\mathrm{ph}}^2) = 0, \qquad m_{\mathrm{ph}}^2 = m^2 + \Sigma(m_{\mathrm{ph}}^2). \tag{2.18}
$$

This equation fixes only the pole position. Away from the pole, the full propagator is not simply $\mathrm{i}/(p^2 - m_{\mathrm{ph}}^2)$; the remaining momentum dependence of $\Sigma(p^2)$ still contributes.

In the on-shell scheme the renormalized mass parameter is chosen to be the physical pole mass. Thus the loop integral is written with m_{ph}:

$$
\Sigma_1(p^2) = \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k\, \frac{1}{(k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)((p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)}. \tag{2.19}
$$

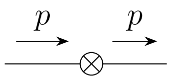
The bare mass is then split as

$$
m_0^2 = m_{\mathrm{ph}}^2 + \delta m^2. \tag{2.20}
$$

In the Lagrangian this means

$$
-\frac{1}{2}m_0^2\phi^2 = -\frac{1}{2}m_{\mathrm{ph}}^2\phi^2 - \frac{1}{2}\delta m^2\phi^2. \tag{2.21}
$$

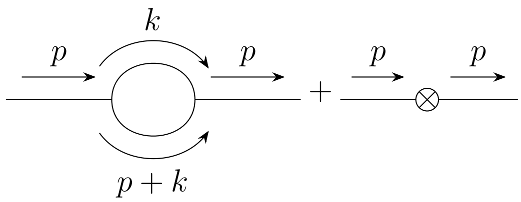
The counterterm vertex is therefore



$$
= -\mathrm{i}\,\delta m^2. \tag{2.22}
$$

The renormalized one-loop self-energy is the sum of the ordinary bubble and the counterterm graph:

$$-\mathrm{i}\Sigma_{1R}(p^2) =$$



$$= -\mathrm{i}\Sigma_1(p^2) - \mathrm{i}\delta m^2. \tag{2.23}$$

Equivalently,

$$\Sigma_{1R}(p^2) = \Sigma_1(p^2) + \delta m^2 = \Sigma_{1,\mathrm{fin}}(p^2) + \big(\Sigma_{1,\mathrm{div}} + \delta m^2\big). \tag{2.24}$$

Choosing

$$\delta m^2 = -\Sigma_{1,\mathrm{div}} \tag{2.25}$$

cancels the divergent part. In the on-shell scheme, the remaining finite part is fixed by the condition that the pole sits at

$$p^2 = m_{\mathrm{ph}}^2. \tag{2.26}$$

A useful way to compute the finite answer is to differentiate with respect to p^2. The counterterm and the subtracted reference integral do not depend on p, so the derivative acts only on the original loop integral:

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= \frac{p^\mu}{2p^2}\frac{\partial \Sigma_1}{\partial p^\mu} \\ &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int \mathrm{d}^4k\,\frac{p\cdot(p+k)}{(k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)((p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)^2}.\end{aligned} \tag{2.27}$$

To evaluate it, use

$$\frac{1}{AB^n} = \int_0^1 \mathrm{d}x\,\frac{nx^{n-1}}{[xB + (1-x)A]^{n+1}}, \tag{2.28}$$

with

$$A = k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad B = (p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad n = 2. \tag{2.29}$$

Then

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k\,\frac{2x\,p\cdot(p+k)}{\big[k^2 + 2x\,p\cdot k + xp^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon\big]^3} \\ &= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k\,\frac{x\,p\cdot(p+k)}{\big[m_{\mathrm{ph}}^2 - k^2 - 2x\,p\cdot k - xp^2 - \mathrm{i}\epsilon\big]^3}.\end{aligned} \tag{2.30}$$

Complete the square by shifting the loop momentum:

$$k' = k + xp, \qquad \Delta(x) = m_{\mathrm{ph}}^2 - p^2 x(1-x). \tag{2.31}$$

The numerator becomes

$$x\,p\cdot(p+k) = x\big[p^2(1-x) + p\cdot k'\big]. \tag{2.32}$$

The term proportional to $p\cdot k'$ is odd in the shifted integration variable and vanishes. Therefore

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{\mathrm{i}g^2}{16\pi^4}\int_0^1 \mathrm{d}x\,x(1-x)\int \mathrm{d}^4k'\,\frac{1}{\big[\Delta(x) - k'^2 - \mathrm{i}\epsilon\big]^3}. \tag{2.33}$$

Using

$$\int \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{1}{(l^2 - \Delta + \mathrm{i}\epsilon)^m} = \frac{\mathrm{i}(-1)^m}{(4\pi)^{d/2}} \frac{\Gamma(m - d/2)}{\Gamma(m)} \Delta^{d/2-m}, \tag{2.34}$$

with $d = 4$ and $m = 3$, gives

$$\begin{aligned} \frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \frac{x(1-x)}{m_{\mathrm{ph}}^2 - p^2 x(1-x)} \\ &= \frac{g^2}{32\pi^2} \frac{\partial}{\partial p^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - p^2 x(1-x)\right]. \end{aligned} \tag{2.35}$$

Integrating back gives

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - p^2 x(1-x)\right] + C. \tag{2.36}$$

The on-shell condition $\Sigma_{1R}(m_{\mathrm{ph}}^2) = 0$ fixes the constant:

$$C = -\frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - m_{\mathrm{ph}}^2 x(1-x)\right]. \tag{2.37}$$

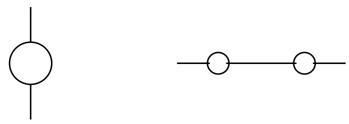
Thus

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[\frac{m_{\mathrm{ph}}^2 - p^2 x(1-x)}{m_{\mathrm{ph}}^2(1 - x + x^2)}\right]. \tag{2.38}$$

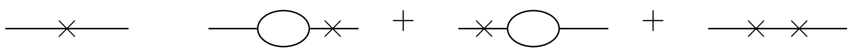
This expression is finite and vanishes at $p^2 = m_{\mathrm{ph}}^2$ by construction.

At higher orders, the same logic must be applied to divergent subgraphs inside larger graphs. If a one-loop bubble appears as a subgraph, the renormalized graph is obtained by adding the corresponding local counterterm graph:

1. The original larger graph may contain a bubble subgraph:



2. Each divergent bubble must be accompanied by a counterterm insertion. The cross denotes the local counterterm:



3. The sum of the original graph and all required counterterm graphs is finite. In this sense the divergent bubble subgraph is replaced by its finite renormalized value $-\mathrm{i}\Sigma_{1R}$.

The important point is locality: whenever a divergent short-distance subgraph appears, its divergent part has the same form as a local counterterm with the same external legs.

1.3 Degree of divergence and required counterterms

The superficial degree of divergence is a quick estimate of how a Feynman integral behaves at large loop momentum. For the one-loop self-energy in d spacetime dimensions,

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.39}$$

At large k, each propagator behaves as $1/k^2$, while the measure contributes $k^{d-1}\mathrm{d}k$. Therefore

$$\mathrm{d}^d k\,\frac{1}{k^2k^2} \sim \mathrm{d}k\,k^{d-5}. \tag{2.40}$$

Equivalently, before the final radial integral, the graph has superficial degree

$$D = d-4. \tag{2.41}$$

Thus

$$d<4 \implies \text{the graph is superficially convergent,} \tag{2.42}$$
$$d=4 \implies \text{the graph is logarithmically divergent,} \tag{2.43}$$
$$d>4 \implies \text{the graph is power divergent by superficial counting.} \tag{2.44}$$

External-momentum derivatives reduce the degree of divergence. Differentiating once gives

$$\frac{\partial\Sigma_1}{\partial p^\mu} = -\frac{\mathrm{i}g^2}{(2\pi)^d}\int \mathrm{d}^d k\,\frac{(p+k)_\mu}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)^2}. \tag{2.45}$$

At large k,

$$\mathrm{d}^d k\,\frac{k}{k^2(k^2)^2} \sim \mathrm{d}k\,k^{d-6}, \tag{2.46}$$

so the degree has been lowered from $d-4$ to $d-5$. If $d=5$, naive power counting gives a logarithm. The leading logarithmic piece is odd under symmetric integration of the loop momentum, so it does not produce a Lorentz-invariant divergence in the self-energy.

Differentiating a second time gives

$$\frac{\partial^2\Sigma_1}{\partial p^\mu\partial p^\nu} = \frac{\mathrm{i}g^2}{(2\pi)^d}\int \mathrm{d}^d k\,\frac{2(p+k)_\mu(p+k)_\nu - g_{\mu\nu}((p+k)^2-m^2)}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)^3}. \tag{2.47}$$

Its superficial degree is

$$D_{\partial^2\Sigma} = d-6. \tag{2.48}$$

In four dimensions, this is finite. When we integrate twice to reconstruct Σ_1, the most general missing polynomial is

$$A + B_\mu p^\mu. \tag{2.49}$$

Since the self-energy is a Lorentz scalar, the linear term is forbidden, so $B_\mu = 0$. The only local ambiguity in $d=4$ is the constant A, which is precisely a mass counterterm.

In six dimensions, the original graph has $D=2$. After enough derivatives make the integral finite, integrating back allows a quadratic local polynomial:

$$A + Bp^2 + C_\mu p^\mu. \tag{2.50}$$

Lorentz invariance again sets $C_\mu = 0$. Therefore in $d=6$ the two-point counterterm must contain both a mass counterterm and a wave-function counterterm. With the sign convention used below, their contribution to the self-energy is

$$\Sigma_{\mathrm{ct}}(p^2) = \delta m^2 - \delta Z\,p^2. \tag{2.51}$$

The corresponding local Lagrangian terms are

$$\mathcal{L}_{\text{ct}} = -\frac{1}{2}\delta Z\, \partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2}\delta m^2 \phi^2. \tag{2.52}$$

The δZ term is called wave-function or field-strength renormalization.

The field-strength factor relates the bare field to the renormalized field:

$$\phi_0 = Z^{1/2}\phi, \qquad \phi = Z^{-1/2}\phi_0. \tag{2.53}$$

Therefore

$$\begin{aligned} G_{2(0)}(p) &= \langle 0|\, \mathrm{T}\{\phi_0(p)\phi_0(-p)\}\, |0\rangle \\ &= Z\, \langle 0|\, \mathrm{T}\{\phi(p)\phi(-p)\}\, |0\rangle = Z G_2(p). \end{aligned} \tag{2.54}$$

The full bare propagator is

$$G_{2(0)}(p^2) = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_{1(0)}(p^2) + \mathrm{i}\epsilon}. \tag{2.55}$$

The physical mass is the pole position:

$$m_{\text{ph}}^2 - m_0^2 - \Sigma_{1(0)}(m_{\text{ph}}^2) = 0, \qquad \Sigma_{1(0)}(m_{\text{ph}}^2) = m_{\text{ph}}^2 - m_0^2. \tag{2.56}$$

Hence

$$p^2 - m_0^2 - \Sigma_{1(0)}(p^2) = p^2 - m_{\text{ph}}^2 - \big[\Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\text{ph}}^2)\big]. \tag{2.57}$$

When only mass renormalization is needed, this motivates

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\text{ph}}^2). \tag{2.58}$$

With a wave-function counterterm included, the general form is

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) + \delta m^2 - \delta Z\, p^2. \tag{2.59}$$

Then the bare and renormalized propagators are related by

$$G_{2(0)}(p^2) = \frac{\mathrm{i}Z}{p^2 - m_{\text{ph}}^2 - \Sigma_{1R}(p^2) + \mathrm{i}\epsilon}. \tag{2.60}$$

The on-shell scheme imposes two conditions. The first fixes the pole:

$$\Sigma_{1R}(m_{\text{ph}}^2) = 0. \tag{2.61}$$

The second fixes the pole residue. Expanding near $p^2 = m_{\text{ph}}^2$,

$$\begin{aligned} \Sigma_{1R}(p^2) &= \Sigma_{1R}(m_{\text{ph}}^2) + (p^2 - m_{\text{ph}}^2)\Sigma_{1R}'(m_{\text{ph}}^2) + O\big((p^2 - m_{\text{ph}}^2)^2\big) \\ &= (p^2 - m_{\text{ph}}^2)\Sigma_{1R}'(m_{\text{ph}}^2) + O\big((p^2 - m_{\text{ph}}^2)^2\big). \end{aligned} \tag{2.62}$$

Thus

$$G_2(p^2) \simeq \frac{\mathrm{i}}{(p^2 - m_{\text{ph}}^2)\big[1 - \Sigma_{1R}'(m_{\text{ph}}^2)\big]}. \tag{2.63}$$

Requiring unit residue gives

$$\Sigma'_{1R}(m^2_{\mathrm{ph}}) = 0. \tag{2.64}$$

For the bare propagator,

$$\begin{aligned} G_{2(0)}(p^2) &\simeq \frac{\mathrm{i}}{(p^2 - m^2_{\mathrm{ph}})\left[1 - \Sigma'_{1(0)}(m^2_{\mathrm{ph}})\right]} \\ &\equiv \frac{\mathrm{i}R_{(0)}}{p^2 - m^2_{\mathrm{ph}}}, \end{aligned} \tag{2.65}$$

so

$$R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m^2_{\mathrm{ph}})}. \tag{2.66}$$

Since $G_{2(0)} = ZG_2$, choosing the renormalized propagator to have unit residue means

$$Z = R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m^2_{\mathrm{ph}})}. \tag{2.67}$$

Finally, rewrite the bare ϕ^3 Lagrangian:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi_0\,\partial^\mu\phi_0 - \frac{1}{2}m_0^2\phi_0^2 - \frac{g_0}{3!}\phi_0^3. \tag{2.68}$$

Using $\phi_0 = Z^{1/2}\phi$,

$$\mathcal{L} = \frac{1}{2}Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}Zm_0^2\phi^2 - \frac{g_0}{3!}Z^{3/2}\phi^3. \tag{2.69}$$

Define

$$Z = 1 - \delta Z, \qquad Zm_0^2 = m^2_{\mathrm{ph}} + \delta m^2, \qquad Z^{3/2}g_0 = g_{\mathrm{ph}} + \delta g. \tag{2.70}$$

Then

$$\begin{aligned} \mathcal{L} = {} & \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2_{\mathrm{ph}}\phi^2 - \frac{g_{\mathrm{ph}}}{3!}\phi^3 \\ & - \frac{1}{2}\delta Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3. \end{aligned} \tag{2.71}$$

The first line supplies the renormalized propagator and vertex. The second line supplies the counterterm vertices needed to cancel divergences order by order.

1.4 Arbitrariness in a renormalization graph

Counterterms are required by finiteness, but finiteness does not uniquely determine their finite parts. This freedom is the origin of different renormalization schemes.

In four-dimensional ϕ^3 theory at one loop, the two-point graph is logarithmically divergent, so a mass counterterm is enough and we may set

$$\delta Z = 0 \tag{2.72}$$

at this order. The renormalized self-energy can then be written as

$$\Sigma_{1R}(p^2) = \Sigma_{1,\mathrm{fin}}(p^2) + \left(\Sigma_{1,\mathrm{div}} + \delta m^2\right). \tag{2.73}$$

The divergent part of δm^2 must cancel $\Sigma_{1,\mathrm{div}}$, but the finite part is still a convention.

In the mass-shell scheme, the convention is fixed by requiring the pole of the full propagator to be at $p^2 = m_{\mathrm{ph}}^2$:

$$\Sigma_{1R}(m_{\mathrm{ph}}^2) = 0. \tag{2.74}$$

Thus

$$0 = \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + \Sigma_{1,\mathrm{div}} + \delta m^2, \tag{2.75}$$

and hence

$$\delta m_{\mathrm{OS}}^2 = -\Sigma_{1,\mathrm{div}} - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2). \tag{2.76}$$

The on-shell renormalized self-energy is

$$\Sigma_{1R}^{\mathrm{OS}}(p^2) = \Sigma_{1,\mathrm{fin}}(p^2) - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2). \tag{2.77}$$

Therefore, through order g^2,

$$G_2(p^2) = \frac{\mathrm{i}}{p^2 - m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{fin}}(p^2) + \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4)}. \tag{2.78}$$

The bare mass is

$$m_0^2 = m_{\mathrm{ph}}^2 + \delta m_{\mathrm{OS}}^2 = m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{div}} - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{2.79}$$

Now choose a different finite part. The only requirement for finiteness is

$$\delta m^2 = -\Sigma_{1,\mathrm{div}} + \text{finite}. \tag{2.80}$$

For example, choose the finite part to be $-g^2 C$:

$$\delta m_C^2 = -\Sigma_{1,\mathrm{div}} - g^2 C, \tag{2.81}$$

where C is an arbitrary finite constant. The renormalized self-energy is then

$$\Sigma_{1R}^{(C)}(p^2, m^2) = \Sigma_{1,\mathrm{fin}}(p^2, m^2) - g^2 C. \tag{2.82}$$

Here m is the renormalized mass parameter in this scheme. It is not automatically the physical pole mass. The physical pole must be found from

$$p^2 - m^2 - \Sigma_{1R}^{(C)}(p^2, m^2) = 0 \tag{2.83}$$

order by order in perturbation theory. The propagator is

$$G_2^{(C)}(p^2) = \frac{\mathrm{i}}{p^2 - m^2 - \Sigma_{1,\mathrm{fin}}(p^2, m^2) + g^2 C + O(g^4)}. \tag{2.84}$$

The bare mass in this scheme is

$$m_0^2 = m^2 + \delta m_C^2 = m^2 - \Sigma_{1,\mathrm{div}} - g^2 C + O(g^4). \tag{2.85}$$

The bare mass is the same object in every scheme. Equating the on-shell expression and the C-scheme expression gives

$$m^2 - \Sigma_{1,\mathrm{div}} - g^2 C = m_{\mathrm{ph}}^2 - \Sigma_{1,\mathrm{div}} - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{2.86}$$

The divergent pieces cancel, leaving

$$m^2 = m_{\mathrm{ph}}^2 + g^2 C - \Sigma_{1,\mathrm{fin}}(m_{\mathrm{ph}}^2) + O(g^4). \tag{2.87}$$

This is a finite redefinition between two renormalization schemes. The parameter m is scheme dependent, while m_{ph} is defined by the pole of the propagator and is observable. Changing the finite part of the counterterm changes the numerical value assigned to the renormalized parameter, but it does not change physical predictions when all quantities are converted consistently.

1.5 Dimensional regularization

Dimensional regularization is another systematic way to handle ultraviolet divergences. Instead of putting a hard cutoff Λ on the loop momentum, we temporarily continue the number of spacetime dimensions from 4 to a complex variable d. The integral is first evaluated in a region of d where it converges, and the result is then analytically continued back to the physical value. In this language, UV divergences appear as poles of Gamma functions.

For the same one-loop two-point function in ϕ^3 theory, we write

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2\mu^{4-d}}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.88}$$

The factor μ^{4-d} is inserted so that the coupling g keeps its four-dimensional mass dimension while d is moved away from 4.

The basic Gaussian integral is

$$\int \mathrm{d}^n x\, \mathrm{e}^{-x^2} = (\sqrt{\pi})^n = \pi^{n/2}. \tag{2.89}$$

After Wick rotation, $k^0 = \mathrm{i}k_E^0$, the Minkowski square becomes $k^2 = -k_E^2$, and the measure contributes one factor of i:

$$\begin{aligned}
\int \mathrm{d}^d k\, \mathrm{e}^{k^2} &= \mathrm{i} \int \mathrm{d}^d k_E\, \mathrm{e}^{-k_E^2} \\
&= \mathrm{i}\left(\int \mathrm{d}k_E^0\, \mathrm{e}^{-(k_E^0)^2}\right)\left(\int \mathrm{d}^{d-1}\vec{k}_E\, \mathrm{e}^{-\vec{k}_E^2}\right) \\
&= \mathrm{i}\sqrt{\pi}\,\pi^{(d-1)/2} = \mathrm{i}\pi^{d/2}.
\end{aligned} \tag{2.90}$$

One way to derive the dimensionally regularized answer is to use Schwinger parameters. The propagator can be represented as

$$\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} = \int_0^\infty \mathrm{d}a\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)}. \tag{2.91}$$

Using one parameter for each propagator gives

$$\begin{aligned}
\Sigma_1 &= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)} \mathrm{e}^{-\mathrm{i}b(m^2 - (p+k)^2 - \mathrm{i}\epsilon)} \\
&= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}[(a+b)m^2 - (a+b)k^2 - bp^2 - 2b\, p\cdot k - \mathrm{i}\epsilon(a+b)]}.
\end{aligned} \tag{2.92}$$

Complete the square by defining

$$k' = k + \frac{b}{a+b} p. \tag{2.93}$$

Then

$$(a+b)k^2 + 2b\, p\cdot k + bp^2 = (a+b)k'^2 + \frac{ab}{a+b} p^2. \tag{2.94}$$

It is convenient to change variables from (a, b) to

$$z = a + b, \qquad x = \frac{a}{a+b}, \qquad a = xz, \qquad b = (1-x)z. \tag{2.95}$$

The Jacobian is

$$\mathrm{d}a\,\mathrm{d}b = \begin{vmatrix} \frac{\partial a}{\partial x} & \frac{\partial a}{\partial z} \\ \frac{\partial b}{\partial x} & \frac{\partial b}{\partial z} \end{vmatrix} \mathrm{d}x\,\mathrm{d}z = \begin{vmatrix} z & x \\ -z & 1-x \end{vmatrix} \mathrm{d}x\,\mathrm{d}z = z\,\mathrm{d}x\,\mathrm{d}z. \tag{2.96}$$

Thus $z \in (0, \infty)$ and $x \in (0, 1)$, and the exponent contains the standard combination

$$\Delta(x) = m^2 - p^2 x(1-x). \tag{2.97}$$

After the Gaussian loop integral and the z integral, one obtains

$$\Sigma_1(p^2, m^2; d) = -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\left[m^2 - p^2x(1-x)\right]^{d/2-2}. \tag{2.98}$$

The same result follows more directly from the Feynman-parameter identity

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x\,\frac{1}{[xB + (1-x)A]^2}. \tag{2.99}$$

With

$$A = k^2 - m^2 + \mathrm{i}\epsilon, \qquad B = (p+k)^2 - m^2 + \mathrm{i}\epsilon, \tag{2.100}$$

we get

$$\begin{aligned} \Sigma_1 &= \frac{\mathrm{i}g^2\mu^{4-d}}{2}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^dk}{(2\pi)^d}\frac{1}{[(k+xp)^2 - \Delta(x) + \mathrm{i}\epsilon]^2} \\ &= -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\Delta(x)^{d/2-2}. \end{aligned} \tag{2.101}$$

The divergence is now entirely in $\Gamma(2 - d/2)$. Near $d = 4$, set

$$d = 4 - \epsilon. \tag{2.102}$$

Then

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \tag{2.103}$$

and

$$\Delta(x)^{d/2-2} = \Delta(x)^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln\Delta(x) + O(\epsilon^2). \tag{2.104}$$

Using also

$$\mu^\epsilon(4\pi)^{\epsilon/2} = 1 + \epsilon\ln\mu + \frac{\epsilon}{2}\ln 4\pi + O(\epsilon^2), \tag{2.105}$$

the unrenormalized self-energy becomes

$$\Sigma_1 = -\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln\Delta(x)\right] + O(\epsilon). \tag{2.106}$$

1.6 Minimal subtraction scheme

As we introduced above, we may have different schemes to choose for renormalization. The on-shell subtraction is especially transparent in dimensional regularization. Define

$$\Sigma_{1R}^{\mathrm{OS}}(p^2) = \Sigma_1(p^2, m_{\mathrm{ph}}^2; d) - \Sigma_1(m_{\mathrm{ph}}^2, m_{\mathrm{ph}}^2; d). \tag{2.107}$$

The two terms have the same pole, so the pole cancels before taking $\epsilon \to 0$. Since

$$\Delta_p(x) = m_{\text{ph}}^2 - p^2 x(1-x), \qquad \Delta_{\text{OS}}(x) = m_{\text{ph}}^2 - m_{\text{ph}}^2 x(1-x) = m_{\text{ph}}^2(1-x+x^2), \tag{2.108}$$

we obtain

$$\Sigma_{1R}^{\text{OS}}(p^2) = \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x \,\ln \frac{m_{\text{ph}}^2 - p^2 x(1-x)}{m_{\text{ph}}^2(1-x+x^2)}. \tag{2.109}$$

This is exactly the finite on-shell result obtained earlier by differentiating with respect to p^2.

But on-shell scheme would have problems when we are dealing with the massless case. So we introduce the **Minimal Subtraction (MS)**, which is a different prescription. It subtracts only the pole part at the physical dimension. In $d = 4 - \epsilon$, the mass counterterm in the MS scheme is chosen as

$$\delta m_{\text{MS}}^2 = \frac{g^2}{32\pi^2}\frac{2}{\epsilon}. \tag{2.110}$$

Therefore

$$\begin{aligned}\Sigma_{1R}^{\text{MS}}(p^2) &= \Sigma_1(p^2) + \delta m_{\text{MS}}^2 \\ &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x \left[\ln\left(\frac{\Delta(x)}{4\pi\mu^2}\right) + \gamma_E\right] + O(\epsilon).\end{aligned} \tag{2.111}$$

This expression is finite, but it is not the on-shell scheme result. In particular, $\Sigma_{1R}^{\text{MS}}(m^2)$ is not required to vanish. The MS scheme removes only the pole; the finite part is chosen by convention.

The scale μ appears because the mass dimension of the coupling changes when the dimension is continued. In d dimensions,

$$[\mathcal{L}] = [m]^d, \qquad [\phi] = [m]^{(d-2)/2}, \tag{2.112}$$

and the interaction

$$\frac{g}{3!}\phi^3 \tag{2.113}$$

implies

$$[g] = [m]^{(6-d)/2}. \tag{2.114}$$

Around four dimensions, g has dimension one, and writing $g_0 = \mu^{(4-d)/2} g$ keeps the renormalized coupling normalized as a four-dimensional parameter. Around six dimensions, where ϕ^3 is classically marginal, the corresponding continuation is

$$g_0 = \mu^{3-d/2}(g + \delta g). \tag{2.115}$$

Now consider the same two-point graph near six dimensions. Set

$$d = 6 - \epsilon, \qquad \Delta(x) = m^2 - p^2 x(1-x). \tag{2.116}$$

The dimensionally regularized result is

$$\Sigma_1 = -\frac{g^2\mu^{6-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \Delta(x)^{d/2-2}. \tag{2.117}$$

Here

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(-1+\frac{\epsilon}{2}\right) = -\frac{2}{\epsilon} + \gamma_E - 1 + O(\epsilon). \tag{2.118}$$

Expanding the remaining factors gives

$$\Sigma_1 = -\frac{g^2}{128\pi^3}\int_0^1 \mathrm{d}x\,\Delta(x)\left[-\frac{2}{\epsilon}+\gamma_E-1+\ln\left(\frac{\Delta(x)}{4\pi\mu^2}\right)\right]+O(\epsilon). \tag{2.119}$$

Since

$$\int_0^1 \mathrm{d}x\,\Delta(x) = m^2-\frac{p^2}{6}, \tag{2.120}$$

the pole part is

$$\Sigma_{1,\text{pole}} = \frac{g^2}{64\pi^3}\frac{1}{\epsilon}\left(m^2-\frac{p^2}{6}\right). \tag{2.121}$$

Thus a mass counterterm alone is no longer sufficient: the divergence contains both a constant and a p^2 term. Writing

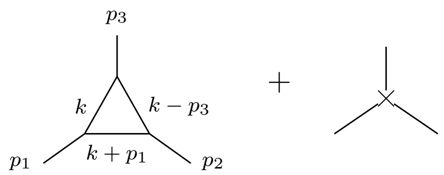
$$\Sigma_{1R} = \Sigma_1+\delta m^2-\delta Z\,p^2, \tag{2.122}$$

minimal subtraction at $d=6-\epsilon$ requires

$$\delta m^2 = -\frac{g^2m^2}{64\pi^3}\frac{1}{\epsilon} = \frac{g^2m^2}{64\pi^3}\frac{1}{d-6}, \tag{2.123}$$

$$\delta Z = -\frac{g^2}{6\cdot 64\pi^3}\frac{1}{\epsilon} = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6}. \tag{2.124}$$

In six dimensions the three-point vertex is also logarithmically divergent. The tree vertex is corrected by the one-loop triangle, and the divergence is canceled by a local three-leg counterterm:



To extract the UV pole, set the external momenta to zero after noting that the superficial divergence is logarithmic. Momentum-dependent terms are less divergent and affect only finite parts. The triangle contribution is then

$$\begin{aligned}\Gamma_1^{(3)} &= (-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k+p_1)^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k-p_2)^2-m^2+\mathrm{i}\epsilon}\\ &\xrightarrow{p_i\to 0}(-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}^3}{(k^2-m^2+\mathrm{i}\epsilon)^3}\\ &= -\frac{\mathrm{i}g^3}{2(4\pi)^3}\left(\frac{2}{\epsilon}+\text{finite}\right).\end{aligned} \tag{2.125}$$

The counterterm vertex is $-\mathrm{i}\delta g$. Requiring the pole to cancel gives

$$-\mathrm{i}\delta g+\left[-\frac{\mathrm{i}g^3}{2(4\pi)^3}\frac{2}{\epsilon}\right]=0, \tag{2.126}$$

so

$$\delta g = -\frac{g^3}{64\pi^3}\frac{1}{\epsilon}\,\mu^{3-d/2} = \frac{g^3}{64\pi^3}\frac{1}{d-6}\,\mu^{3-d/2}. \tag{2.127}$$

This is the vertex counterterm in minimal subtraction near six dimensions.

1.7 Two loop graph in massless ϕ^3 theory

The on-shell scheme is natural for a massive particle, because the physical mass is defined by an isolated pole at $p^2 = m_{\text{ph}}^2$. For a strictly massless interacting theory this prescription becomes delicate. The reason is not a new ultraviolet problem; it is an infrared problem caused by taking the subtraction point to the massless pole.

Recall the one-loop on-shell mass counterterm in dimensional regularization $d = 4 - \epsilon$:

$$\delta m_{\text{OS}}^2 = -\Sigma_1(m_{\text{ph}}^2, m_{\text{ph}}^2) = \frac{g^2\mu^\epsilon}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\left[m_{\text{ph}}^2(1-x+x^2)\right]^{d/2-2}, \tag{2.128}$$

with the approximation formulas:

$$\Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad A^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln A + O(\epsilon^2). \tag{2.129}$$

These gives

$$\delta m_{\text{OS}}^2 = \frac{g^2}{32\pi^2}\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln m_{\text{ph}}^2 - \int_0^1 \mathrm{d}x\,\ln\big(1-x+x^2\big)\right] + O(\epsilon). \tag{2.130}$$

The pole $2/\epsilon$ is the ultraviolet divergence. The separate term

$$-\ln m_{\text{ph}}^2 \tag{2.131}$$

blows up when $m_{\text{ph}} \to 0$. This is an infrared divergence. It appears because the on-shell subtraction point has been moved to $p^2 = 0$, where a massless field has long-range propagation.

Minimal subtraction does not require an on-shell subtraction point. Setting $m = 0$ after MS subtraction gives

$$\begin{aligned}\Sigma_{1R}^{\text{MS}}(p^2)\big|_{m=0} &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\left[\ln\left(\frac{-p^2x(1-x)}{4\pi\mu^2}\right) + \gamma_E\right] \\ &= \frac{g^2}{32\pi^2}\left[\ln\left(\frac{-p^2}{4\pi\mu^2}\right) + \gamma_E - 2\right].\end{aligned} \tag{2.132}$$

This expression is meaningful for $p^2 \neq 0$, but it is still singular as $p^2 \to 0$. Therefore the failure of the mass-shell scheme in the strictly massless case is an infrared statement: near the massless point, the full propagator is not described by the same simple isolated massive-particle pole.

For perturbation theory it is useful to separate the Lagrangian into the free part, the basic interaction, and the counterterms:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_b + \mathcal{L}_{ct} \tag{2.133}$$

with

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \tag{2.134}$$

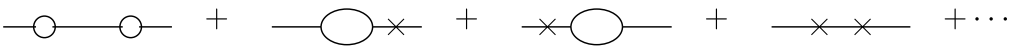
$$\mathcal{L}_b = -\frac{g}{3!}\phi^3, \tag{2.135}$$

and

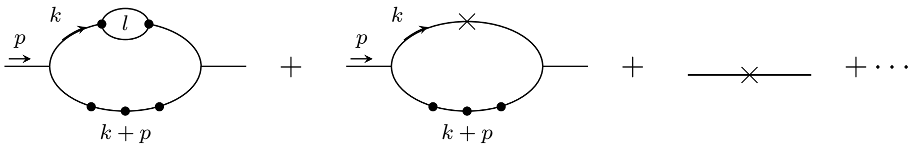
$$\mathcal{L}_{ct} = -\frac{1}{2}\delta Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3 - \delta h\,\phi. \tag{2.136}$$

The term δh cancels tadpoles, namely one-point divergences. At order g^4 we must draw not only the genuine two-loop graphs. We must also draw graphs in which a lower-order divergent subgraph has already been replaced by its local counterterm. The relevant classes are:

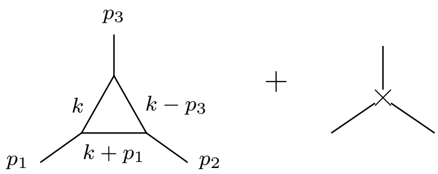
1. Iterated one-loop self-energy insertions on the same propagator line:



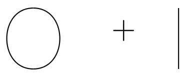
2. A two-loop self-energy graph with a one-loop two-point subdivergence:



3. A one-loop three-point vertex correction and its vertex counterterm:

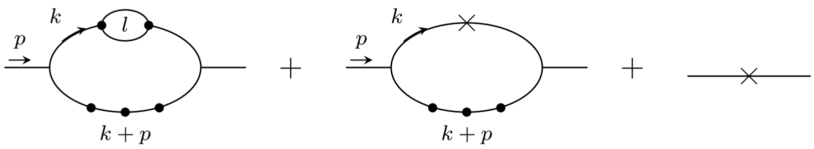


Each cross stands for a local counterterm. The rule is simple but important: every divergent subgraph needs its own subtraction before the remaining overall divergence can be removed. Tadpoles are not shown in the main list, because they are local one-point divergences and are canceled by the $\delta h\,\phi$ term:



A pole multiplying a non-polynomial function such as $\ln(-p^2)$ is called a non-local subdivergence. It cannot be canceled by a local term in the Lagrangian. After all subdivergences have been subtracted, the remaining overall divergence is local: in momentum space it is a polynomial in the external momenta.

Now we consider the two-loop self-energy in $d = 6$. The diagram that matters for subdivergences is the graph in which a one-loop two-point subgraph is inserted into an internal line. We must include three contributions:



(a) (b) (c)

Graph (a) is the genuine two-loop graph. Graph (b) replaces the divergent one-loop subgraph inside (a) by the lower-order counterterm. Graph (c) is the remaining overall local two-point counterterm. For a two-point function the cross abbreviates the local insertion $\delta m^2 - \delta Z\, p^2$; in the massless calculation below the relevant overall counterterm is the wave-function part. The sum of all three graphs is finite.

$$\Sigma_{2a} = (\mathrm{i}g\mu^{3-d/2})^4 \frac{1}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{l^2 - m^2 + \mathrm{i}\epsilon}$$
$$\times \frac{\mathrm{i}}{(k-l)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(k+p)^2 - m^2 + \mathrm{i}\epsilon}. \tag{2.137}$$

The two factors of $1/(k^2-m^2)$ are the two propagator segments on the line that contains the one-loop subgraph. For convenience, we now take the massless case:

$$\Sigma_{2a} = \frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{d}^d l}{(2\pi)^d}\frac{1}{(k^2+\mathrm{i}\epsilon)^2(l^2+\mathrm{i}\epsilon)((k-l)^2+\mathrm{i}\epsilon)((k+p)^2+\mathrm{i}\epsilon)}. \tag{2.138}$$

After Wick rotation, we have

$$\begin{aligned}
\Sigma_{2a} &= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{\mathrm{d}^d l_E}{(2\pi)^d}\frac{1}{[l_E^2(k_E-l_E)^2][(k_E^2)^2(k_E+p_E)^2]}\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\left(\int \frac{\mathrm{d}^d l_E}{(2\pi)^d}\frac{1}{l_E^2(k_E-l_E)^2}\right)\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\frac{\Gamma(2-d/2)}{(2\pi)^d}\int_0^1 \mathrm{d}x\,[-k_E^2 x(1-x)]^{d/2-2}\\
&= -\frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{1}{(k_E^2)^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(-k_E^2)^{d/2-2}\\
&= \frac{\mathrm{i}g^4}{2^{13}\pi^9}(16\pi^3\mu^4)^{3-d/2}\frac{\Gamma(2-d/2)\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}\int \mathrm{d}^d k_E\,\frac{1}{(-k_E^2)^{4-d/2}(k_E+p_E)^2}.
\end{aligned} \tag{2.139}$$

For the integral, one can introduce a Feynman parameter

$$\frac{1}{A^{4-d/2}B} = \frac{\Gamma(5-d/2)}{\Gamma(4-d/2)}\int_0^1 \mathrm{d}x\,\frac{x^{3-d/2}}{[Ax+B(1-x)]^{5-d/2}}. \tag{2.140}$$

Thus, the result is

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\frac{\Gamma(2-d/2)\Gamma(5-d)\Gamma(\frac{d}{2}-1)^3\Gamma(d-4)}{\Gamma(d-2)\Gamma(4-d/2)\Gamma(3d/2-5)} \tag{2.141}$$

$$\equiv \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\Gamma(2-d/2)\Gamma(5-d)K(d) \tag{2.142}$$

Here $\Gamma(2-d/2)$ is the pole of the one-loop subgraph, while $\Gamma(5-d)$ is the remaining overall pole. The massless formula also has an infrared singularity as $p^2\to 0$; this is separate from the ultraviolet subdivergence.

Now expand near six dimensions. To keep track of the non-local momentum dependence, define

$$d = 6-\epsilon, \qquad L_p = \ln\left(\frac{-p^2}{4\pi\mu^2}\right). \tag{2.143}$$

The singular factors are

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(-1+\frac{\epsilon}{2}\right) = -\frac{2}{\epsilon}+O(1), \tag{2.144}$$

$$\Gamma(5-d) = \Gamma(-1+\epsilon) = -\frac{1}{\epsilon}+O(1), \tag{2.145}$$

$$\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6} = 1-\epsilon L_p+\frac{\epsilon^2}{2}L_p^2+O(\epsilon^3). \tag{2.146}$$

Therefore graph (a) has the pole structure

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{1}{(d-6)^2}+\frac{L_p+c_a}{d-6}+\frac{1}{2}L_p^2+c_a' L_p+c_a''+O(d-6)\right]. \tag{2.147}$$

The constants c_a, c'_a, c''_a are not important for the present point. The important term is $L_p/(d-6)$: it is a pole multiplying a non-polynomial function of the external momentum. A local counterterm cannot cancel such a term directly. It must be canceled by graph (b), which subtracts the divergent one-loop subgraph before the remaining overall divergence is removed.

For $m = 0$, the one-loop two-point counterterm in the subgraph is the wave-function counterterm

$$\delta_1 Z = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6} + O(g^4). \tag{2.148}$$

Because the counterterm in graph (b) sits on the line carrying momentum k, its Feynman rule contributes a factor proportional to $\delta_1 Z\, k^2$:

$$\begin{aligned}
\Sigma_{2b} &= (-\mathrm{i}g\mu^{3-d/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+p)^2-m^2}\delta_1 Z\cdot k^2 && (2.149)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z \int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{k_E^2}{(k_E^2)^2(k_E+p_E)^2} && (2.150)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z \frac{\mathrm{i}\pi^{d/2}}{(2\pi)^d}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(p^2)^{d/2-2} && (2.151)\\
&= \left(\frac{g^2}{64\pi^3}\right)\left(\frac{p^2}{6}\right)\frac{\Gamma(2-d/2)\Gamma^2(\frac{d}{2}-1)}{(d-6)\Gamma(d-2)}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d/2-3} && (2.152)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left(-\frac{2}{\epsilon}+\gamma_E-1\right)\frac{1}{\epsilon}\left(1-\frac{\epsilon}{2}L_p+\frac{\epsilon^2}{8}L_p^2\right) && (2.153)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left[-\frac{2}{\epsilon^2}+\frac{\gamma_E-1+L_p}{\epsilon}-\frac{1}{2}(\gamma_E-1)L_p-\frac{1}{4}L_p^2\right] && (2.154)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[-\frac{2}{(d-6)^2}+\frac{-L_p+\text{const.}}{d-6}-\frac{1}{4}L_p^2+\text{const.}\,L_p+O(d-6)\right]. && (2.155)
\end{aligned}$$

Adding graphs (a) and (b), the pole multiplying L_p cancels:

$$\begin{aligned}
&\Sigma_{2a}+\Sigma_{2b}\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{-1}{(d-6)^2}+\frac{1}{d-6}\text{Const.}+\frac{1}{4}\ln^2\frac{-p^2}{4\pi\mu^2}+\text{Const.}\ln\frac{-p^2}{4\pi\mu^2}+\text{const.}+O(d-6)\right]
\end{aligned} \tag{2.156}$$

The non-local pole has disappeared:

$$\frac{1}{d-6}\left(\ln\frac{-p^2}{4\pi\mu^2}+\text{Const.}\right)\to\frac{1}{d-6}\text{const.} \tag{2.157}$$

If this term survived, canceling it would require a non-local counterterm schematically of the form

$$\phi(x)\ln(\Box)\phi(x) \tag{2.158}$$

for the subdivergence $\ln(p^2)/(d-6)$, which is unacceptable in a local renormalizable Lagrangian. Since the non-local pole has been canceled by graph (b), the remaining divergent part is local and can be canceled by graph (c). The two-loop wave-function counterterm is fixed by

$$-p^2\delta_2 Z = -(\Sigma_{2a}+\Sigma_{2b})_{\text{local pole}}. \tag{2.159}$$

Equivalently, in MS notation,

$$\delta_2 Z = \left(\frac{g^2}{64\pi^3}\right)^2 \frac{1}{36}\left[-\frac{1}{(d-6)^2} + \frac{c_Z}{d-6}\right], \tag{2.160}$$

where c_Z depends on the precise subtraction convention.

Thus, the renormalized result is

$$\begin{aligned} \Sigma_{2R}^{\mathrm{MS}} &= \Sigma_{2a} + \Sigma_{2b} - p^2\delta_2 Z \\ &= \left(\frac{g^2}{64\pi^3}\right)^2 \frac{p^2}{36}\left[\frac{1}{4}L_p^2 + c_1 L_p + c_0\right]. \end{aligned} \tag{2.161}$$

1.8 External-momentum derivatives and locality of counterterms

We now explain why the remaining overall counterterm of a 1PI graph must be a polynomial in the external momenta. This is the basic locality statement behind renormalization: after all subdivergences have been subtracted, the only ultraviolet divergence left in a graph is local.

For the two-loop graph above in $d = 6$, the superficial degree of the overall divergence is $D = 2$. Each derivative with respect to the external momentum lowers the degree by one. Therefore three external-momentum derivatives should remove the overall divergence. To make the bookkeeping clean, use the shorthand

$$\partial_\mu^p \equiv \frac{\partial}{\partial p^\mu}. \tag{2.162}$$

We also isolate the one-loop subgraph as

$$\Pi(k) = \int \mathrm{d}^d l\, \frac{1}{l^2(k-l)^2}. \tag{2.163}$$

Then the two-loop graph has the schematic form

$$\Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \frac{1}{(k^2)^2\,(k+p)^2}, \tag{2.164}$$

$$\partial_\mu^p \partial_\nu^p \partial_\rho^p \Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \partial_\mu^p \partial_\nu^p \partial_\rho^p \left[\frac{1}{(k^2)^2\,(k+p)^2}\right]. \tag{2.165}$$

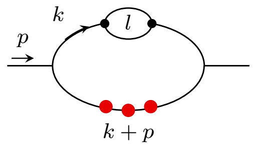
The derivative acts only on propagators through which the external momentum flows. In the graph above this is the line with momentum $k+p$:

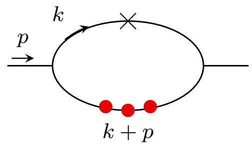
$$\partial_\mu^p \frac{1}{(k+p)^2 - m^2} = -\frac{2(k+p)_\mu}{\left[(k+p)^2 - m^2\right]^2}. \tag{2.166}$$

Thus differentiating a propagator produces one extra propagator power and a numerator proportional to the momentum through that line. In graph language we denote this bookkeeping operation by adding a marked insertion on the differentiated line:

$$\partial_\mu^p(\underline{\qquad}) \quad \longrightarrow \quad \underline{\qquad}. \tag{2.167}$$

The three red dots below mean that the external-momentum dependent propagator has been differentiated three times. They are not new interaction vertices; they only remind us where the powers of momentum in the numerator come from. The differentiated versions of graphs (a) and (b) are shown below.

(a''') 

(b''') 

The claim is slightly subtle. The graph (a''') by itself still contains the same one-loop subdivergence in the l loop. The finite object is the subtracted combination

$$(a''') + (b'''). \tag{2.168}$$

After this subtraction, the third derivative of the full graph is ultraviolet finite. Equivalently, the divergent part of the un-differentiated graph must obey

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \Sigma_{2,\mathrm{div}}(p) = 0. \tag{2.169}$$

Therefore $\Sigma_{2,\mathrm{div}}(p)$ can be at most a quadratic polynomial in the external momentum:

$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + B_\mu(d)p^\mu + C(d), \tag{2.170}$$
$$B_\mu(d) = 0 \qquad \text{by Lorentz invariance}, \tag{2.171}$$
$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + C(d). \tag{2.172}$$

The term $A(d)p^2$ is canceled by a wave-function counterterm. The constant $C(d)$ is a mass counterterm; in the massless MS example discussed above, the displayed overall divergence is proportional to p^2, so the relevant graph (c) is the δZ counterterm.

Let us check explicitly why the subtracted third derivative is finite. There are three ultraviolet regions to consider. If k and l become large together, the original overall degree is $D = 2$, and three derivatives lower it to $D = -1$, so this region is convergent. If $l \to \infty$ while k is fixed, the divergence is precisely the one-loop subgraph and is canceled by (b'''). The only region that is easy to miss is the nested region

$$l \gg k \gg |p|. \tag{2.173}$$

In this region the loop momentum l is much larger than the momentum k entering the subgraph, so the inner loop can be expanded in powers of k/l. Define

$$I(k) = \int \mathrm{d}^6 l \, \frac{1}{l^2 (l+k)^2}. \tag{2.174}$$

Here we keep $d = 6$, because this is the case where the two-loop graph has the subdivergence we want to subtract. The useful expansion is

$$(l+k)^2 = l^2 + 2l\cdot k + k^2 = l^2 \left(1 + \frac{2l\cdot k + k^2}{l^2}\right), \tag{2.175}$$
$$\frac{1}{(l+k)^2} = \frac{1}{l^2}\left[1 - \frac{2l\cdot k + k^2}{l^2} + \left(\frac{2l\cdot k + k^2}{l^2}\right)^2 + \cdots\right]. \tag{2.176}$$

This is the same as the Taylor expansion of the subgraph in its external momentum k:

$$I(k) = I(0) + k^\mu \left.\frac{\partial I}{\partial k^\mu}\right|_{k=0} + \frac{1}{2}k^\mu k^\nu \left.\frac{\partial^2 I}{\partial k^\mu \partial k^\nu}\right|_{k=0} + \cdots . \tag{2.177}$$

Now look term by term. In six dimensions $\mathrm{d}^6 l = \Omega_5 l^5 \mathrm{d}l$, and angular symmetry gives $\langle (l\cdot k)^2\rangle = l^2k^2/6$. The first three terms behave as

$$I_{(0)}(k) = \int \mathrm{d}^6 l \, \frac{1}{l^4} \sim \Omega_5 \int^{\Lambda} \mathrm{d}l \, l \sim \Lambda^2, \tag{2.178}$$

$$I_{(1)}(k) = -\int \mathrm{d}^6 l \, \frac{2l\cdot k}{l^6} = -2k_\mu \int \mathrm{d}^6 l \, \frac{l^\mu}{l^6} = 0, \tag{2.179}$$

$$\begin{aligned} I_{(2)}(k) &= \int \mathrm{d}^6 l \left(-\frac{k^2}{l^6} + \frac{4(l\cdot k)^2}{l^8} \right) \\ &\sim \Omega_5 \int^{\Lambda} \mathrm{d}l \, l^5 \left(-\frac{k^2}{l^6} + \frac{4}{l^8}\frac{l^2k^2}{6} \right) \\ &= -\frac{\Omega_5 k^2}{3} \int^{\Lambda} \frac{\mathrm{d}l}{l} \sim -\frac{\Omega_5 k^2}{3} \ln \Lambda. \end{aligned} \tag{2.180}$$

Thus $I_{(0)}$ is quadratically divergent, $I_{(1)}$ vanishes by oddness, and $I_{(2)}$ is logarithmically divergent.

Thus the divergent part of the subgraph is local in k. More explicitly, the divergent part can be written as a polynomial in k; the appearance of that polynomial depends on the regulator:

$$\begin{aligned} I^{(\Lambda)}_{\mathrm{div}}(k) &= a_0\Lambda^2 + a_2 k^2 \ln \frac{\Lambda^2}{\mu^2}, && \text{hard cutoff,} \\ I^{(\mathrm{DR})}_{\mathrm{div}}(k) &= \frac{b_2 k^2}{d-6}, && \text{dimensional regularization near } d = 6. \end{aligned} \tag{2.181}$$

The constants a_0, a_2, b_2 are regulator dependent. What matters is not their numerical value here, but the fact that the divergent part is a local polynomial: a constant plus a term proportional to k^2. The counterterm in graph (b) removes exactly these local divergent pieces. After subtraction, the large-k behavior of the inner subgraph is at worst

$$I_R(k) = I(k) - I_{\mathrm{ct}}(k) = O\big(k^2 \ln k^2\big). \tag{2.182}$$

Terms of order k^3 and higher in the expansion of the subgraph are already finite in the l loop, so they do not introduce new subdivergences.

Finally combine this with the differentiated outer line. For $k \gg |p|$,

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \frac{1}{(k+p)^2} = O\left(\frac{1}{k^5}\right). \tag{2.183}$$

The worst large-k behavior of the subtracted differentiated two-loop graph is therefore obtained by multiplying the six-dimensional measure, the renormalized subgraph, the two k^2 propagators on the upper line, and the differentiated $k+p$ propagator:

$$\int^{\infty} \mathrm{d}k \, k^5 \left(k^2 \ln k^2\right) \frac{1}{(k^2)^2} \frac{1}{k^5} \sim \int^{\infty} \mathrm{d}k \, \frac{\ln k}{k^2} < \infty. \tag{2.184}$$

which is finite. This is the concrete reason why the non-local subdivergence must cancel between graphs (a) and (b), leaving only a local polynomial counterterm.

1.9 Composite operators

The new object in this subsection is a product of fields at nearby points, for example $\phi(x)\phi(y)$ when $x-y$ is small. Such a product is more singular than an ordinary field, because two quantum fields are being multiplied at almost the same spacetime point. The useful statement is that, at short distances, this product can be replaced by a sum of local operators at one point:

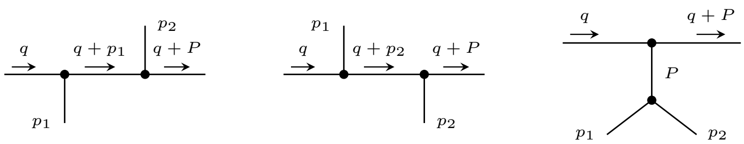
$$\phi(x)\phi(y) \sim C_1(x-y)\mathbb{1}+C_{\phi^2}(x-y)\left[\frac{1}{2}\phi^2(y)\right]+C_\phi(x-y)[\phi(y)]+C^\mu_{\partial\phi}(x-y)[\partial_\mu\phi(y)]+\dots \tag{2.185}$$

This is the operator product expansion, or OPE. The bracket $[A]$ means that the local operator has been renormalized. The functions C_i are ordinary functions, not operators; they are called Wilson coefficients. A Wilson coefficient contains the short-distance, high-momentum part of a graph, while the matrix element of the local operator contains the soft long-distance part.

We now see this mechanism in ϕ^3 theory. The large momentum q flows through the pair of fields that are being brought close together, whereas p_1 and p_2 are held fixed. In the labels of the figures,

$$P = p_1 + p_2, \qquad q^2 \gg p_1^2,\ p_2^2,\ P^2,\ m^2. \tag{2.186}$$

The relevant contributions to the four-point function are



(a) hard (b) hard (c) soft

The words "hard" and "soft" only describe which subgraph carries the large momentum. In graphs (a) and (b), the three propagators along the upper line are all hard. In graph (c), only the two upper propagators are hard; the lower part, which carries momentum P, remains soft and will be identified with a matrix element of a local operator.

The expansion used repeatedly below is the large-q expansion of a propagator. If p is fixed while q is taken large, then

$$\frac{1}{(q+p)^2-m^2} = \frac{1}{q^2}\left[1-\frac{2q\cdot p+p^2-m^2}{q^2}+O\left(\frac{1}{q^2}\right)\right]. \tag{2.187}$$

Equivalently, the small parameters are p/q and m/q. Only propagators carrying the hard momentum are expanded. The soft denominators are kept exact:

$$D_1 = p_1^2-m^2, \qquad D_2 = p_2^2-m^2, \qquad D_P = P^2-m^2. \tag{2.188}$$

$$(a) = (-\mathrm{i}g)^2\frac{\mathrm{i}}{q^2-m^2}\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{(q+p_1)^2-m^2}\frac{\mathrm{i}}{(q+P)^2-m^2}\frac{\mathrm{i}}{D_2} \tag{2.189}$$

$$\xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \tag{2.190}$$

$$(b) \xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \tag{2.191}$$

$$(a) + (b) \simeq 2\mathrm{i}g^2 \left(\frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2} \right) \frac{1}{(q^2)^3}, \tag{2.192}$$

$$(c) = (-\mathrm{i}g)^2 \frac{\mathrm{i}}{q^2 - m^2} \frac{\mathrm{i}}{(q+P)^2 - m^2} \frac{\mathrm{i}}{D_P} \frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2}. \tag{2.193}$$

For graph (c), the hard expansion contains two hard propagators instead of three:

$$\begin{aligned}(c) = &-\frac{\mathrm{i}g^2}{(q^2)^2} \left[1 - \frac{2q \cdot P}{q^2} + \frac{2m^2 - P^2}{q^2} + \frac{4(q \cdot P)^2}{(q^2)^2} + \cdots \right] \\ &\times \frac{1}{(P^2 - m^2)(p_1^2 - m^2)(p_2^2 - m^2)}, \qquad P = p_1 + p_2. \end{aligned} \tag{2.194}$$

The first important point is that the soft factor in (a)+(b) is exactly the lowest-order matrix element of the local operator $\frac{1}{2}\phi^2(0)$:

$$\begin{aligned} M_{\phi^2}(p_1, p_2) &\equiv \langle 0 | \, \mathrm{T}() \{ \tilde{\phi}(p_1) \tilde{\phi}(p_2) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}} \\ &= \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}. \end{aligned} \tag{2.195}$$

The factor $\frac{1}{2}$ removes the double counting of the two identical contractions. In the connected part,

$$\begin{aligned} &\langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}} \\ &\quad = \frac{1}{2} D(x) D(y) + \frac{1}{2} D(y) D(x) = D(x) D(y). \end{aligned} \tag{2.196}$$

The contraction in which the two fields inside $\phi^2(0)$ contract with each other gives a tadpole proportional to $D(0)$; it is not part of the connected graph displayed here and is removed by operator renormalization. In momentum space:

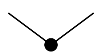
$$\begin{aligned} &\int \mathrm{d}^4x \, \mathrm{d}^4y \, \mathrm{e}^{-\mathrm{i}p_1 \cdot x - \mathrm{i}p_2 \cdot y} D(x) D(y) \\ &\quad = \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}, \end{aligned} \tag{2.197}$$

$$M_{\phi^2}(p_1, p_2) = \int \mathrm{d}^4x \, \mathrm{d}^4y \, \mathrm{e}^{-\mathrm{i}p_1 \cdot x - \mathrm{i}p_2 \cdot y} \langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle_{0,\mathrm{conn}}. \tag{2.198}$$

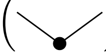
Here $\phi(0)$ is not Fourier transformed, because it is the local operator inserted at the origin. The same rule follows directly from the path integral:

$$\langle 0 | \, \mathrm{T}() \{ \phi(x) \phi(y) \frac{1}{2} \phi^2(0) \} \, | 0 \rangle = N \int [\mathrm{d}A] \, A(x) A(y) \frac{1}{2} A^2(0) \mathrm{e}^{\mathrm{i}S[A]}. \tag{2.199}$$

Here ϕ denotes the quantum operator, while A is the classical field inside the path integral. At lowest order, the insertion of $\frac{1}{2}\phi^2(0)$ therefore gives one local black-dot vertex with two scalar lines:



This insertion $\frac{1}{2}\phi^2(0)$ is our first example of a composite operator. Therefore

$$
\begin{aligned}
(a)+(b) &\simeq \underbrace{\frac{2\mathrm{i}g^2}{(q^2)^3}}_{C_{\phi^2}(q)} \underbrace{M_{\phi^2}(p_1,p_2)}_{\text{soft matrix element}} \\
&= \frac{2\mathrm{i}g^2}{(q^2)^3} \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\mathrm{conn}} \\
&= \frac{2\mathrm{i}g^2}{(q^2)^3}\Big(
\end{aligned}
$$

$$\Big). \tag{2.200}$$

Graph (c) is slightly different. The soft part is not the matrix element of $\frac{1}{2}\phi^2$; it is the matrix element of the elementary field ϕ between the vacuum and two external scalar legs. This matrix element first appears at order g:

$$
\begin{aligned}
&\langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle_{O(g)} \\
&\quad= \langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\int \mathrm{d}^4w\,\frac{-\mathrm{i}g}{3!}\phi^3(w)\}\,|0\rangle_0 \\
&\quad= -\mathrm{i}g\int \mathrm{d}^4w\, D(x-w)D(y-w)D(z-w).
\end{aligned}
\tag{2.201}
$$

We use

$$
D(u-v) = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\mathrm{e}^{\mathrm{i}k\cdot(u-v)}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}. \tag{2.202}
$$

In momentum space, with $P = p_1+p_2$, define

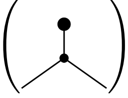
$$
\begin{aligned}
M_\phi(x;p_1,p_2) &\equiv \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle_{O(g)} \\
&= \int \mathrm{d}^4y\mathrm{d}^4z\,\mathrm{e}^{-\mathrm{i}p_1\cdot y-\mathrm{i}p_2\cdot z}\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}P\cdot w}D(x-w) \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}\,\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2-m^2+\mathrm{i}\epsilon}\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}(P+k)\cdot w} \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\frac{\mathrm{i}}{P^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{-\mathrm{i}P\cdot x}.
\end{aligned}
\tag{2.203}
$$

The factor $\mathrm{e}^{-\mathrm{i}P\cdot x}$ is the important part. Once the soft subgraph is represented by a local operator at $x=0$, powers of P in the hard coefficient become derivatives acting on that local operator:

$$
\partial_\mu M_\phi(x;p_1,p_2)\big|_{x=0} = -\mathrm{i}P_\mu M_\phi(0;p_1,p_2), \qquad P_\mu \to \mathrm{i}\partial_\mu, \qquad P^2 \to -\Box. \tag{2.204}
$$

$$
(c) \sim \frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2q\cdot P}{q^2}+\frac{2m^2-P^2}{q^2}+\frac{4(q\cdot P)^2}{(q^2)^2}+\cdots\right]M_\phi(0;p_1,p_2) \tag{2.205}
$$

$$
= \frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2\mathrm{i}q^\mu\partial_\mu}{q^2}+\frac{2m^2+\Box}{q^2}-\frac{4q^\mu q^\nu\partial_\mu\partial_\nu}{(q^2)^2}+\cdots\right]\langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle\Big|_{x=0} \tag{2.206}
$$

$$
= \frac{\mathrm{i}g}{(q^2)^2}\sum_n c_n(q,\partial)\Bigg(
$$

$$\Bigg). \tag{2.207}$$

Putting the two types of terms together gives the OPE form of this four-point function:

$$\begin{aligned} &\langle 0|\,\mathrm{T}()\{\tilde{\phi}(q)\phi(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle \\ &\quad \sim \sum_i C_i(q)\,\langle 0|\,\mathrm{T}()\{O_i(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle\,. \end{aligned} \tag{2.208}$$

In this example the first few local operators are

$$O_1 = \frac{1}{2}\phi^2, \qquad \text{from graphs (a) and (b),} \tag{2.209}$$

$$O_2 = \phi, \qquad \text{from the leading term of graph (c),} \tag{2.210}$$

$$O_3 = \partial_\mu\phi,\ \Box\phi, \ldots, \qquad \text{from higher terms in graph (c).} \tag{2.211}$$

The lesson is the separation of scales: the large momentum q is stored in the Wilson coefficients $C_i(q)$, while the dependence on the soft external momenta is stored in matrix elements of local operators.

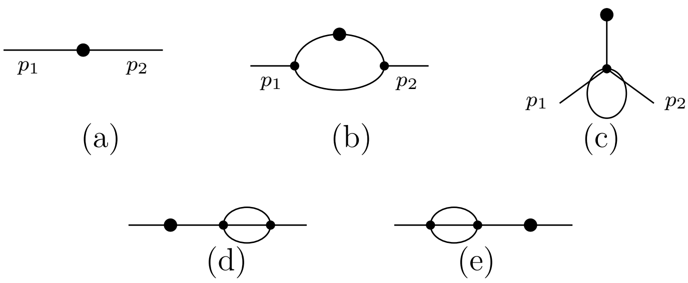
1.10 Renormalization of composite operators

The previous subsection used $\frac{1}{2}\phi^2(0)$ as a local insertion. This is not yet a complete definition of the operator. When the fields inside ϕ^2 approach interaction vertices, new ultraviolet divergences can occur at the insertion point itself. These divergences are not removed by renormalizing only the parameters in the Lagrangian; they require a renormalization of the composite operator.

Consider ϕ^3 theory in $d = 6 - \epsilon$ dimensions and the Green function with one insertion of $\frac{1}{2}\phi^2$:

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle \tag{2.212}$$

The connected graphs up to order g^2 are shown below. The black dot is the insertion of $\frac{1}{2}\phi^2$. A crossed mark, when it appears later, denotes a local counterterm.



We now Fourier transform the two ordinary external fields, while the composite operator remains localized at the point z:

$$\begin{aligned} G &= \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(z)\}\,|0\rangle \\ &= \int \mathrm{d}^d x\,\mathrm{d}^d y\,\mathrm{e}^{\mathrm{i}(p_1\cdot x + p_2\cdot y)}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle\,. \end{aligned} \tag{2.213}$$

The lowest-order graph (a) is just the insertion vertex connected to the two external propagators. We use the shorthand

$$S_i = \frac{\mathrm{i}}{p_i^2 - m^2 + \mathrm{i}\epsilon}, \qquad S_{12} = S_1 S_2. \tag{2.214}$$

$$G_a = S_{12}. \tag{2.215}$$

For the order-g^2 graphs, the large-k behavior is

$$G_b \sim \int \mathrm{d}^6 k \, \frac{1}{k^2}\frac{1}{k^2}\frac{1}{k^2} \sim \int^\Lambda \frac{\mathrm{d}k}{k}, \qquad \text{logarithmically divergent,} \tag{2.216}$$

$$G_c,\ G_d,\ G_e \sim \int \mathrm{d}^6 k \, \frac{1}{k^4} \sim \int^\Lambda \mathrm{d}k\, k, \qquad \text{quadratically divergent by power counting.} \tag{2.217}$$

Graphs (d) and (e) are ordinary self-energy corrections on the external lines. Their divergences are cancelled by the usual wave-function and mass counterterms. After ordinary field and parameter renormalization has been performed, they do not give a new counterterm for $[\phi^2]$.

Graphs (b) and (c) are different. Their short-distance divergence sits at the composite-operator insertion itself, so the bare operator ϕ^2 is not enough. We must add local counterterms to define the finite operator $[\phi^2]$.

$$G_b = S_{12}(-\mathrm{i}g)^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_1 - k)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_2 + k)^2 - m^2 + \mathrm{i}\epsilon}. \tag{2.218}$$

After the continuation $g \to g\mu^{(6-d)/2}$, the loop part is

$$\begin{aligned} G_b = S_{12}&\left(\frac{\mathrm{i}g^2\mu^{6-d}}{(2\pi)^d}\right) \\ &\times \int \mathrm{d}^d k \, \frac{1}{(k^2 - m^2)[(p_1 - k)^2 - m^2][(p_2 + k)^2 - m^2]}. \end{aligned} \tag{2.219}$$

$$\begin{aligned} &\frac{1}{(k^2 - m^2)[(p_1 - k)^2 - m^2][(p_2 + k)^2 - m^2]} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(1-x-y)(k^2 - m^2) + x((p_1 - k)^2 - m^2) + y((p_2 + k)^2 - m^2)\right]^{-3} \end{aligned} \tag{2.220}$$

$$\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k^2 - m^2 - 2(xp_1 - yp_2)\cdot k + xp_1^2 + yp_2^2\right]^{-3}. \tag{2.221}$$

Completing the square with $k' = k - xp_1 + yp_2$ and $P = p_1 + p_2$ gives

$$\begin{aligned} &2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(k - xp_1 + yp_2)^2 - (xp_1 - yp_2)^2 + xp_1^2 + yp_2^2 - m^2\right]^{-3} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k'^2 - \Delta_b(x,y)\right]^{-3}, \end{aligned} \tag{2.222}$$

$$\Delta_b(x,y) = m^2 - p_1^2 x(1-x-y) - p_2^2 y(1-x-y) - P^2 xy. \tag{2.223}$$

After Wick rotation, the momentum integral is

$$\begin{aligned} &\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \int \mathrm{d}^d k_E \, \frac{2}{(k_E^2 + \Delta_b)^3} \\ &\quad = \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \frac{(2\pi)^d}{(4\pi)^{d/2}} \Gamma\!\left(3 - \frac{d}{2}\right) \Delta_b^{d/2-3}. \end{aligned} \tag{2.224}$$

Therefore

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3}. \tag{2.225}$$

The pole is especially transparent if we write $d = 6-\epsilon$. Since

$$\Gamma\left(3-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y = \frac{1}{2}, \tag{2.226}$$

the divergent part of graph (b) is

$$G_b^{\text{pole}} = S_{12}\frac{g^2}{64\pi^3}\frac{1}{\epsilon} = -S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{2.227}$$

Thus the operator counterterm that cancels this pole is local and proportional to the same two-leg insertion $\frac{1}{2}\phi^2$.

For graph (c), use the same P and set

$$D_{12P} = (p_1^2 - m^2)(p_2^2 - m^2)(P^2 - m^2). \tag{2.228}$$

Then

$$G_c = (-\mathrm{i}g\mu^{\frac{6-d}{2}})^2 \frac{-\mathrm{i}}{D_{12P}}\frac{1}{2!}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+P)^2-m^2} \tag{2.229}$$

$$= \frac{-\mathrm{i}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k}{(2\pi)^d}\left[(k+xP)^2 - \Delta_c(x)\right]^{-2},$$

$$\Delta_c(x) = m^2 - P^2 x(1-x) \tag{2.230}$$

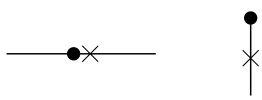
$$= \frac{\frac{1}{2}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{1}{(k_E^2+\Delta_c(x))^2} \tag{2.231}$$

$$= \frac{\frac{1}{2}g^2(\mu^2)^{3-d/2}}{D_{12P}}\frac{1}{(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \Delta_c(x)^{d/2-2} \tag{2.232}$$

$$= \frac{g^2}{D_{12P}}\frac{1}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{2.233}$$

$$G_c = \frac{-g\mu^{3-d/2}}{D_{12P}}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x\, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{2.234}$$

G_b and G_c both diverge, so the bare composite operator ϕ^2 does not define finite Green functions by itself. For the OPE we need a local operator with the same quantum numbers and a finite insertion. In the MS scheme, the required counterterms are obtained by replacing each divergent short-distance loop by a local vertex:



For graph (b), the pole part is

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right)\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3} \tag{2.235}$$

$$= S_{12}\frac{g^2}{64\pi^3}\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[\frac{2}{\epsilon} - \gamma_E - \ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right] + O(\epsilon) \tag{2.236}$$

$$= -S_{12}\frac{g^2}{64\pi^3(d-6)} + \text{finite}. \tag{2.237}$$

Therefore the corresponding MS counterterm insertion is

$$G_{b,\text{ct}} = S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{2.238}$$

For graph (c), the pole part is

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}}, \tag{2.239}$$

where $\Delta_c(x) = m^2 - P^2 x(1-x)$, $\qquad P = p_1 + p_2$, so

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\left(-\frac{2}{\epsilon}\right)\int_0^1 \mathrm{d}x\, \Delta_c(x) + \text{finite} \tag{2.240}$$

$$= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right) + \text{finite}. \tag{2.241}$$

Thus

$$G_{c,\text{ct}} = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\left\{\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)\right\}. \tag{2.242}$$

This is exactly what is produced by inserting the local operator

$$\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\phi. \tag{2.243}$$

Indeed, in momentum space $\Box$ acting on the local field gives $-P^2$, so $m^2+\Box/6$ becomes $m^2-P^2/6$. Throughout this subsection $d=6-\epsilon$, so $1/(d-6) = -1/\epsilon$; this is the source of the signs in the MS counterterms.

Thus, the renormalized value at $d=6$ is

$$R(G_b) = \frac{\mathrm{i}^2}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3}\left[-\frac{\gamma_E}{2} - \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y\, \ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right], \tag{2.244}$$

$$R(G_c) = \frac{-g}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g}{128\pi^3}\left[(\gamma_E-1)\left(m^2-\frac{P^2}{6}\right) + \int_0^1 \mathrm{d}x\, \Delta_c(x)\ln\frac{\Delta_c(x)}{4\pi\mu^2}\right]. \tag{2.245}$$

The counterterms can now be read as local operator insertions. First, the two-leg insertion satisfies

$$\langle 0|\,\mathrm{T}()\{\tilde\phi(p_1)\tilde\phi(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\text{conn}} = S_{12}. \tag{2.246}$$

Therefore the counterterm from graph (b) is equivalently

$$G_{b,\mathrm{ct}} = \frac{g^2}{64\pi^3(d-6)} \langle 0| \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\} |0\rangle_{0,\mathrm{conn}} . \tag{2.247}$$

Second, the one-leg operator insertion satisfies

$$\langle 0| \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(0)\} |0\rangle_{O(g)} = -\frac{g}{D_{12P}}, \tag{2.248}$$

$$\langle 0| \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\} |0\rangle_{O(g)} = -\frac{g}{D_{12P}}\left(m^2 - \frac{P^2}{6}\right). \tag{2.249}$$

Therefore the counterterm from graph (c) is equivalently

$$G_{c,\mathrm{ct}} = \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0| \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\} |0\rangle_{O(g)} . \tag{2.250}$$

Graphs (d) and (e) have already been handled by the usual field and mass counterterms. The remaining new information is therefore the renormalization of the insertion itself:

$$\begin{aligned} &G_a + R(G_b) + R(G_c) + R(G_d) + R(G_e) \\ &= \left(1 + \frac{g^2}{64\pi^3(d-6)}\right) \langle 0| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\} |0\rangle \\ &\quad + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\} |0\rangle + \text{higher orders.} \end{aligned} \tag{2.251}$$

Thus the finite insertion is generated by the renormalized composite operator

$$\frac{1}{2}[\phi^2] \equiv \left(1 + \frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2 + \frac{1}{6}\Box\right)\phi + \text{higher orders} \tag{2.252}$$

$$= \left(1 + \frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)}m^2\phi + \frac{g\mu^{d/2-3}}{6\cdot 64\pi^3(d-6)}\Box\phi + \text{higher orders.} \tag{2.253}$$

The brackets remind us that this is a renormalized local operator, not the bare product of two fields at the same point. To all orders, the same structure is written as

$$\frac{1}{2}[\phi^2] = Z_a\frac{1}{2}\phi^2 + \mu^{d/2-3}Z_b m^2\phi + \mu^{d/2-3}Z_c\Box\phi. \tag{2.254}$$

The constants Z_a, Z_b, Z_c contain only short-distance information; in minimal subtraction they are series in g and poles in $d-6$. The reason that only these operators appear is the same power-counting logic used before: counterterms must be local, and an operator can mix only with operators whose dimension is not larger than its own and whose quantum numbers match. In six-dimensional ϕ^3 theory, the allowed operators with the right dimension and quantum numbers are ϕ^2, $m^2\phi$, and $\Box\phi$. After ordinary field and mass renormalization has been performed, the remaining operator mixing can be summarized as

$$\begin{pmatrix} \frac{1}{2}[\phi^2] \\ \phi \\ \Box\phi \end{pmatrix} = \begin{pmatrix} Z_a & \mu^{d/2-3}Z_b m^2 & \mu^{d/2-3}Z_c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\phi^2 \\ \phi \\ \Box\phi \end{pmatrix}. \tag{2.255}$$

2. Renormalization in ϕ^4-theory

Firstly, we briefly recall some basic idea about renormalization. For counting the divergent of a theory, one may use the superficial degree of divergence:

$$D = \text{power of } k \text{ in numerator} - \text{power of } k \text{ in denominator}. \tag{2.256}$$

Suppose the Loop integrals in $d = 4$ space-time dimension have superficial degree of divergence D, which can be easily seen from the directly calculating:

$$\int^\Lambda \mathrm{d}^4k \frac{k^{a-4}}{k^b} = \int^\Lambda k^3 \,\mathrm{d}k \,\mathrm{d}\Omega \,\frac{k^{a-b}}{k^4} \sim \int^\Lambda \mathrm{d}k \,\frac{k^{a-b}}{k} = \begin{cases} \Lambda^D & \text{for } D = a - b \neq 0 \\ \ln \Lambda & \text{for } D = 0 \end{cases}, \tag{2.257}$$

where Λ is the cut-off. But we would like to use some more general form to think about it. As an example, considering the ϕ^n-theory

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{n!}\phi^n, \tag{2.258}$$

in which the propagators have the form of $(k^2 - m^2)^{-1}$, vertices are $\sim \lambda$ that connect n lines. So we can define So in this case, we can

1. $L = P - V + 1$
   - L: loops (loop integral with d^4k)
   - P: internal propagators (propagator with $(k^2)^{-1}$)
   - V: vertices (vertices with energy-momentum conservation $\delta(\sum p_i)$)
   - $+1$: global conservation of energy-momentum
2. $nV = 2P + N$
   - n: number of lines at each vertex
   - P: internal propagators
   - N: external lines (or external propagator)

So for each loop, the power of k in numerator is 4, which is contributed by an integral measurement $\sim \mathrm{d}^4k$, and the power of k in denominator is contributed by the propagator. Therefore, the superficial degree of divergence is

$$D = dL - 2P; \quad d = 4. \tag{2.259}$$

Thus, base on the relation we introduced above, we have

$$\begin{aligned} D &= 4(P - V + 1) - 2P \\ &= 2P - 4V + 4 \\ &= (n-4)V - N + 4. \end{aligned} \tag{2.260}$$

Thus, we find D is independent of P, but only depends on N for the case $n = 4$. But for a more general n in $\frac{\lambda}{n!}\phi^n$, we have

- $n = 2$: $\implies$ free theory (included in the mass term);

- $n = 3$: $D = 4 - N - V \implies$ **super renormalizable**
- $n = 4$: $D = -N + 4 \implies$ **renormalizable**
- $n > 4$: $D = (n-4)V - N + 4 \implies$ **non-renormalizable**

For a theory that the signature in front of V is "$-$", we call it as **super renormalizable theory** because the divergence only appear in some leading order terms, with respect the order getting higher, the diagram will finally be convergent, in which $D < 0$. *So in the super renormalizable theory only contains finite number of divergent degree*, which means only need finite counterterms. Similarly, for a theory that calls **renormalizable**, it is because it has no relation with V, which means *no matter how many orders we are counting, the divergent degrees are the same!* Although in this case, it contains infinite divergent diagrams, we can still by re-define the "physical" parameter to include the divergence. In the end, it is quite obvious to understand why we call the last situation as **non-renormalizable**. Because the pre-signature of V is positive, which means with the order getting higher, the divergent will be keep increasing, *so we need infinite parameter to cancel the divergence*, which in most of cases are technically impossible.

Another case is QED, we have two type of propagators, but they all contribute $\sim k^{-2}$

- N_γ: external photon lines;
- N_e: external electron lines;
- P_e: electron propagators;
- P_γ: photon propagators;
- $V = 2P_\gamma + N_\gamma = \frac{1}{2}(2P_e + N_e)$: number of vertices;
- $L = P_e + P_\gamma - V + 1$: number of the loops.

$$\begin{aligned} D &= 4L - P_e - 2P_\gamma \\ &= 4(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma \\ &= 3P_e + 2P_\gamma - 4V + 4 \\ &= 4 - N_\gamma - \frac{3}{2}N_e. \end{aligned} \tag{2.261}$$

So as we discussed above, since D is non-related with V, so QED is a renormalizable theory.

2.1 Analysis of dimensions in view of renormalization

Now we consider another way to analyze the renormalizability. We use the dimension of mass as a standard. So

$$[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1} = [\text{Mass}]^{-1}. \tag{2.262}$$

We require the dimension for the action $[S] = 0$, thus

$$[\mathrm{d}^4x] = -4, \quad [\mathcal{L}] = 4, \quad [\partial_\mu] = 1, \quad [\phi] = 1. \tag{2.263}$$

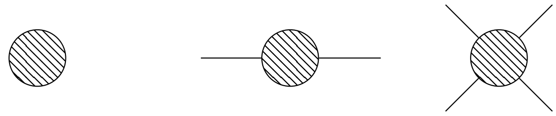
The last one is because $[\partial_\mu\phi\partial^\mu\phi] = 4$. The coupling constant $[\lambda]$ in $\frac{\lambda}{n!}\phi^n$ then has the dimension

$$[\lambda] = 4 - n. \tag{2.264}$$

Take a sharp look! This number, $4-n$, is nothing but the prefactor of superficial degree of divergence in ϕ^n theory, i.e.,

$$D = 4 - N - \underbrace{(4-n)}_{[\lambda]} V. \tag{2.265}$$

Thus, base on the discussion we did above, we find $[\lambda] < 0$ corresponds to a non-renormalizable theory. As an example, we consider the ϕ^4-theory. So $n = 4$ brings the superficial degree of divergence is $D = 4 - N \geq 0$ for degree up to $N = 4$, which is a renormalizable theory.



$D = 4$(vacuum) $D = 2$ (self energy) $D = 0$

We can also do the same thing for QED. For convenience, we rewrite the QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(\mathrm{i}\partial\!\!\!/ - m)\psi + eA_\mu\bar{\psi}\gamma^\mu\psi. \tag{2.266}$$

Similarly , since the action S is dimensionless, so

$$[\psi] = \frac{3}{2}, \quad [A_\mu] = 1. \tag{2.267}$$

Thus again, we can then see the coupling constant e is dimensionless, i.e., $[e] = 0$, which just corresponds to a renormalizable theory.

2.2 Renormalized perturbation theory

Now we still use ϕ^4-theory as an example to show the renormalization procedure. The bare Lagrangian is

$$\mathcal{L}_{\text{bare}} = \frac{1}{2}(\partial_\mu\phi_0)(\partial^\mu\phi_0) - \frac{1}{2}m_0^2\phi_0^2 + \frac{1}{4!}\lambda_0\phi_0^4 \tag{2.268}$$

where the corner mark "0" represent the bare parameters, i.e., λ_0, m_0, and ϕ_0. This Lagrangian is nothing special, it just the original Lagrangian that we used in perturbation theory. Now we just give it a different name to separate with the renormalized Lagrangian that we will discuss later. The name "bare" represent this quantity is not actually the physical quantity that we observe because the bare parameters are actually infinity. These parameters appears only for cancel the infinity in the loop integral. The rest of this canceling is then the physical quantity, which is the renormalized quantity. This explanation shows the core idea of why do we need renormalization, i.e., the physical part of graphs needs to be observable, so we have to find a well-defined quantity that suitable for any order. But in this case, we may have to accept that many "constants" we thought unchanged will start to running with the higher energy scale being considered. And this will be described by the renormalization group equation.

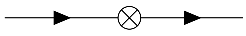
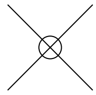
The relation of the renormalized quantities with the bare quantities are

$$\phi_0 = \sqrt{Z}\phi_r, \quad Zm_0^2 = m^2 + \delta m^2, \quad Z^2\lambda_0 = \lambda + \delta\lambda, \tag{2.269}$$

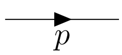
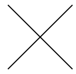
where $Z = 1 + \delta Z$. And δZ, δm^2 and $\delta\lambda$ are called the **counterterms**. So the bare Lagrangian can be written as

$$\begin{aligned}
\mathcal{L}_{bare} &= \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{count}} \\
&= \underbrace{\frac{1}{2}(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}m^2\phi_r^2 + \frac{\lambda}{4!}\phi_r^4}_{\mathcal{L}_{\text{ren}}} + \underbrace{\frac{1}{2}\delta Z(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}\delta m^2\phi_r^2 + \frac{1}{4!}\delta\lambda\phi_r^4}_{\mathcal{L}_{\text{counter term}}}.
\end{aligned} \tag{2.270}$$

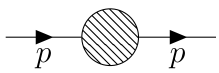
The counterterms also have corresponding Feynman rules

 $= \mathrm{i}(p^2\delta Z - \delta m^2);$  $= -\mathrm{i}\delta\lambda. \qquad (2.271)$

And the Feynman rules of renormalized term is

 $= \dfrac{\mathrm{i}}{p^2 - m^2};$  $= -\mathrm{i}\lambda. \qquad (2.272)$

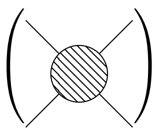
The renormalization condition is usually called the **scheme**. The on-shell scheme for ϕ_r, m in two-point function is



$$
\begin{aligned}
&= \frac{i}{p^2 - m_0^2 - \Sigma(p^2)} && (2.273)\\
&= \frac{\mathrm{i}}{p^2 - m_{\text{phy}}^2} + \underbrace{A + B(p^2 - m_{\text{phy}}^2) + C(p^2 - m_{\text{phy}}^2)^2 + \cdots}_{\text{Regular Terms}}, && (2.274)
\end{aligned}
$$

where th $\Sigma(p^2)$ is the self-energy correction. The main point of this scheme is requiring the residue at the pole $p^2 = m_{\text{phys}}^2$ to be normalized to 1. The mission of the counterterms is to absorb UV divergences and ensure the self-energy $\Sigma(p^2)$ satisfies the specific renormalization conditions, rather than cancelling the regular terms.

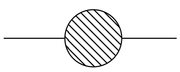
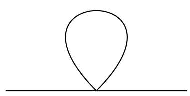
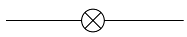
Now for the four-point functions, we impose a similar renormalization condition on the scattering amplitude:

 $_{\text{Amp}} = \mathrm{i}\lambda, \qquad (2.275)$

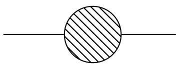
at the threshold $s = 4m^2$ and $t = u = 0$, which corresponds to the case of zero momentum transfer. The Mandelstam variables are defined as usual: $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, and $u = (p_1 - p_4)^2$.This condition, together with the two-point function conditions mentioned earlier, defines the on-shell renormalization scheme. To implement this in practice, we follow these steps:

1. Calculate the divergent diagrams: Evaluate all Feynman diagrams contributing to the n-point functions at a given order in perturbation theory. A suitable regularization scheme (e.g., dimensional regularization) must be chosen to handle the UV divergences.
2. Determine the counterterms: Extract the specific values of the counterterms δm^2, δZ, and $\delta\lambda$ by requiring that the renormalized Green's functions satisfy the chosen renormalization conditions (such as the one in (2.275)).

To conclude this section, we will briefly do the renormalization process for the 1-loop for ϕ^4-theory. Firstly, we expand the two point function until λ^2:

 $=$ $+ \Big($  $+$  $\Big) + O(\lambda^2), \qquad (2.276)$

where the first term is just the normal propagator, the second and the third is the 1-loop graph and the counterterm. Thus, we have



$$= \frac{\mathrm{i}}{p^2 - m^2} - \frac{\mathrm{i}}{2}\lambda \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2} + \mathrm{i}(p^2 \delta Z - \delta m^2) + O(\lambda^2). \tag{2.277}$$

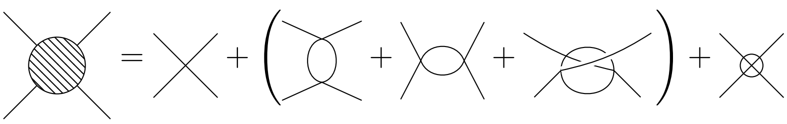
We can by directly calculating the tadpole integral to separate the divergent part, such that to determine the expression for the counterterms. As a result, we only shows the final expression that you should get:

$$\delta Z = 0 \tag{2.278}$$

$$\delta m^2 = -\frac{\lambda}{2} \frac{\Gamma(1 - d/2)}{(4\pi)^{d/2} (m^2)^{1-d/2}}. \tag{2.279}$$

If we take $d = 4 - 2\epsilon$, the theory is renormalizable, so the form of the counterterms are suitable for any order.

The 4-point function also can be expanded into 1-loop order, but since there are two vertices, so the next to next leading order we have is λ^3:



$$+ O(\lambda^3) \tag{2.280}$$

$$= -\mathrm{i}\lambda + (-\mathrm{i}\lambda)^2 \left(\mathrm{i}V(s) + \mathrm{i}V(t) + \mathrm{i}V(u)\right) - \mathrm{i}\delta\lambda + O(\lambda^3) \tag{2.281}$$

The counterterm $\delta\lambda$ is determined by requiring the total scattering amplitude to satisfy the renormalization condition at the chosen subtraction point. According to the Feynman rules, the total amplitude at 1-loop order is the sum of the tree-level vertex, the loop corrections (s, t, and u channels), and the counterterm vertex. For each $\mathrm{i}V(p^2)$, we have

$$\mathrm{i}V(p^2) = \frac{\mathrm{i}}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2} \frac{\mathrm{i}}{(k+p)^2 - m^2} \tag{2.282}$$

$$= -\frac{\mathrm{i}}{2} \frac{1}{(4\pi)^{d/2}} \int_0^1 \mathrm{d}x \, \frac{\Gamma(2 - d/2)}{(m^2 - x(1-x)p^2)^{2-d/2}}, \tag{2.283}$$

where x is the Feynman parameter. By taking $d = 4 - 2\epsilon$, we can separate the divergent part too:

$$\mathrm{i}V(p^2) = -\frac{\mathrm{i}}{32\pi^2} \int_0^1 \mathrm{d}x \left(\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) - \ln\big(m^2 - x(1-x)p^2\big) + O(\epsilon) \right). \tag{2.284}$$

To satisfy the on-shell condition defined in (2.26), we require the amplitude to be exactly $-i\lambda$ at the threshold $s = 4m^2, t = u = 0$. This implies that the loop contributions and the counterterm must cancel each other out at this specific kinematic point:

$$(-i\lambda)^2 \big[\mathrm{i}V(4m^2) + \mathrm{i}V(0) + \mathrm{i}V(0)\big] - \mathrm{i}\delta\lambda = 0 \tag{2.285}$$

Solving for $\delta\lambda$ yields the expression for the 1-loop counterterm in ϕ^4 theory:

$$\delta\lambda = -\lambda^2 \left[V(4m^2) + 2V(0)\right], \tag{2.286}$$

Therefore, the divergence $1/\epsilon$ pole contained in $V(p^2)$ is exactly absorbed by $\delta\lambda$, rendering the final physical amplitude finite. This ensures that the coupling constant λ corresponds to the strength of the interaction measured at the specific energy scale $s = 4m^2$.

3. Renormalization of QED

3.1 The bare theory and the meaning of renormalization

Quantum electrodynamics is the quantum field theory of a Dirac field ψ interacting with a $U(1)$ gauge field A_μ. The classical gauge-invariant part of the Lagrangian is

$$\mathcal{L}_{\rm QED} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\big(\mathrm{i}\not{\partial} - m\big)\psi - e\bar{\psi}\gamma^\mu\psi A_\mu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{2.287}$$

To quantize the photon field perturbatively we must fix a gauge. In a general covariant gauge,

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{2.288}$$

For the explicit one-loop calculations below we use Feynman gauge, $\xi = 1$, because it gives the simplest photon propagator. Since QED is an abelian gauge theory, the Faddeev–Popov ghosts decouple and do not contribute to QED loop diagrams.

The bare Lagrangian is written in terms of bare fields and bare parameters:

$$\mathcal{L}_0 = -\frac{1}{4}F_{0\mu\nu}F_0^{\mu\nu} - \frac{1}{2\xi_0}(\partial_\mu A_0^\mu)^2 + \bar{\psi}_0\big(\mathrm{i}\not{\partial} - m_0\big)\psi_0 - e_0\bar{\psi}_0\gamma^\mu\psi_0 A_{0\mu}. \tag{2.289}$$

Dimensional regularization is used with

$$d = 4 - \epsilon, \qquad \frac{1}{\bar{\epsilon}} \equiv \frac{2}{\epsilon} - \gamma_E + \ln 4\pi. \tag{2.290}$$

The factor $2/\epsilon$ appears because we use $d = 4-\epsilon$ rather than the alternative convention $d = 4-2\epsilon$.

The renormalized fields and parameters are defined by

$$\psi_0 = Z_2^{1/2}\psi, \qquad A_0^\mu = Z_3^{1/2}A^\mu, \qquad m_0 = Z_m m, \qquad e_0 = \mu^{\epsilon/2}Z_e e. \tag{2.291}$$

The scale μ is inserted so that the renormalized charge e remains dimensionless in $d \neq 4$. Substituting these definitions into the bare Lagrangian and separating the renormalized part from the counterterms gives

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \bar{\psi}\big(\mathrm{i}\not{\partial} - m\big)\psi - e\mu^{\epsilon/2}\bar{\psi}\gamma^\mu\psi A_\mu + \mathcal{L}_{\rm ct}, \tag{2.292}$$

$$\mathcal{L}_{\rm ct} = -\frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\big(\mathrm{i}\delta_2\not{\partial} - \delta_m m\big)\psi - e\mu^{\epsilon/2}\delta_1\bar{\psi}\gamma^\mu\psi A_\mu. \tag{2.293}$$

Here

$$\delta_2 = Z_2 - 1, \qquad \delta_3 = Z_3 - 1, \qquad \delta_m = Z_2 Z_m - 1, \qquad \delta_1 = Z_1 - 1. \tag{2.294}$$

It is useful to introduce Z_1 by

$$Z_1 = Z_e Z_2 Z_3^{1/2}. \tag{2.295}$$

This notation means that the coefficient of the interaction $e\bar{\psi}\gamma^\mu\psi A_\mu$ is renormalized by Z_1, while the electron and photon fields are renormalized by Z_2 and Z_3.

The renormalized Feynman rules in Feynman gauge are

$$\text{electron propagator:} \qquad S_F(p) = \frac{\mathrm{i}(\not{p} + m)}{p^2 - m^2 + \mathrm{i}\epsilon}, \tag{2.296}$$

$$\text{photon propagator:} \qquad D_{\mu\nu}(k) = \frac{-\mathrm{i}g_{\mu\nu}}{k^2 + \mathrm{i}\epsilon}, \tag{2.297}$$

$$\text{QED vertex:} \qquad V^\mu = -\mathrm{i}e\mu^{\epsilon/2}\gamma^\mu. \tag{2.298}$$

The counterterm Feynman rules are

$$
\begin{aligned}
&\text{electron two-point counterterm:} && \mathrm{i}\big(\delta_2 \not{p} - \delta_m m\big), && (2.299)\\
&\text{photon two-point counterterm:} && \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big), && (2.300)\\
&\text{vertex counterterm:} && -\mathrm{i}e\mu^{\epsilon/2}\delta_1\gamma^\mu. && (2.301)
\end{aligned}
$$

3.2 Power counting and the allowed counterterms

Before computing integrals, we should know which Green functions can be divergent. Let a QED graph have E_ψ external electron lines and E_A external photon lines. If I_ψ and I_A are the numbers of internal electron and photon propagators, and V is the number of interaction vertices, then

$$2I_\psi + E_\psi = 2V, \qquad 2I_A + E_A = V, \qquad L = I_\psi + I_A - V + 1. \tag{2.302}$$

Each loop integral contributes four powers of momentum, each internal electron propagator contributes one inverse power, and each photon propagator contributes two inverse powers. Therefore the superficial degree of divergence is

$$
\begin{aligned}
D &= 4L - I_\psi - 2I_A \\
&= 4(I_\psi + I_A - V + 1) - I_\psi - 2I_A \\
&= 3I_\psi + 2I_A - 4V + 4 \\
&= 4 - \frac{3}{2}E_\psi - E_A.
\end{aligned} \tag{2.303}
$$

The superficially divergent cases are therefore

$$
\begin{aligned}
&E_\psi = 2,\ E_A = 0: && D = 1, && \text{electron self-energy}, && (2.304)\\
&E_\psi = 0,\ E_A = 2: && D = 2, && \text{photon self-energy}, && (2.305)\\
&E_\psi = 2,\ E_A = 1: && D = 0, && \text{electron-photon vertex}. && (2.306)
\end{aligned}
$$

Other cases are either forbidden or made finite by symmetries. For example, diagrams with an odd number of external photons vanish by Furry's theorem. The four-photon amplitude has $D = 0$, but gauge invariance does not allow a local counterterm of the form A^4 in the QED Lagrangian, and the one-loop light-by-light amplitude is finite after the gauge-invariant sum of diagrams is taken.

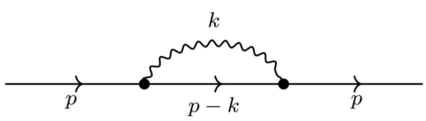
Thus the most general local counterterms needed for QED have exactly the same form as the terms already displayed in $\mathcal{L}_{\text{ct}}$:

$$\bar{\psi}\mathrm{i}\not{\partial}\psi, \qquad m\bar{\psi}\psi, \qquad F_{\mu\nu}F^{\mu\nu}, \qquad e\bar{\psi}\gamma^\mu\psi A_\mu. \tag{2.307}$$

This is the concrete meaning of renormalizability in QED.

3.3 Electron self-energy

The one-loop correction to the electron propagator is



We define the electron self-energy $\Sigma(\not{p})$ by writing the full inverse propagator as

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}). \tag{2.308}$$

In Feynman gauge the one-loop integral is

$$-\mathrm{i}\Sigma(\not{p}) = (-\mathrm{i}e\mu^{\epsilon/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \gamma^\mu \frac{\mathrm{i}(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + \mathrm{i}\epsilon} \gamma^\nu \frac{-\mathrm{i}g_{\mu\nu}}{k^2 + \mathrm{i}\epsilon}. \tag{2.309}$$

The numerator simplifies by the Dirac identities

$$\gamma^\mu \gamma^\alpha \gamma_\mu = (2-d)\gamma^\alpha, \qquad \gamma^\mu \gamma_\mu = d. \tag{2.310}$$

Hence

$$\gamma^\mu (\not{p} - \not{k} + m)\gamma_\mu = (2-d)(\not{p} - \not{k}) + dm. \tag{2.311}$$

To combine the denominators, use

$$\frac{1}{k^2[(p-k)^2 - m^2]} = \int_0^1 \mathrm{d}x \, \frac{1}{\left[k^2 - 2xp\cdot k + x(p^2 - m^2)\right]^2}. \tag{2.312}$$

Complete the square with

$$\ell = k - xp, \qquad \Delta_e(x) = xm^2 - x(1-x)p^2. \tag{2.313}$$

Then the denominator becomes $(\ell^2 - \Delta_e)^2$, and the numerator is

$$(2-d)(\not{p} - \not{k}) + dm = (2-d)\big((1-x)\not{p} - \not{\ell}\big) + dm. \tag{2.314}$$

The term linear in ℓ integrates to zero by symmetry. The divergent part therefore comes from

$$\begin{aligned} \Sigma(\not{p})_{\text{div}} &= \frac{e^2}{16\pi^2}\frac{1}{\epsilon}\int_0^1 \mathrm{d}x \left[(2-d)(1-x)\not{p} + dm\right]_{d=4} \\ &= \frac{e^2}{16\pi^2}\frac{1}{\epsilon}\big[-\not{p} + 4m\big]. \end{aligned} \tag{2.315}$$

This form is important: the divergence is local and has only two possible Dirac structures, $\not{p}$ and m. Therefore it can be canceled by the two electron counterterms in $\mathcal{L}_{\text{ct}}$.

The counterterm contribution to the inverse propagator is

$$\mathrm{i}\big(\delta_2 \not{p} - \delta_m m\big). \tag{2.316}$$

Since the inverse propagator contains $-\Sigma(\not{p})$, the divergent part of

$$-\Sigma(\not{p}) + \delta_2 \not{p} - \delta_m m \tag{2.317}$$

must vanish. This gives

$$\delta_2^{\text{MS}} = -\frac{e^2}{16\pi^2}\frac{1}{\epsilon}, \tag{2.318}$$

$$\delta_m^{\text{MS}} = -\frac{4e^2}{16\pi^2}\frac{1}{\epsilon}. \tag{2.319}$$

The parameter δ_m is the coefficient multiplying $m\bar{\psi}\psi$ inside $Z_2 Z_m$. The mass renormalization constant itself is obtained from

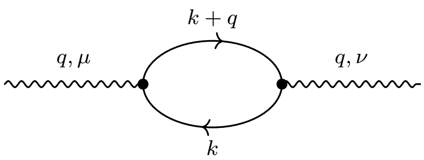
$$Z_2 Z_m = 1 + \delta_m, \qquad Z_2 = 1 + \delta_2. \tag{2.320}$$

To first order in e^2,

$$\begin{aligned} Z_m &= \frac{1+\delta_m}{1+\delta_2} = 1 + \delta_m - \delta_2 + O(e^4) \\ &= 1 - \frac{3e^2}{16\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4). \end{aligned} \tag{2.321}$$

3.4 Photon self-energy and transversality

The one-loop photon self-energy is the fermion loop



The minus sign for a closed fermion loop is essential. The amplitude is

$$\mathrm{i}\Pi^{\mu\nu}(q) = -(-\mathrm{i}e\mu^{\epsilon/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\mathrm{tr}\left[\gamma^\mu \frac{\mathrm{i}(\not{k}+m)}{k^2-m^2+\mathrm{i}\epsilon}\gamma^\nu \frac{\mathrm{i}(\not{k}+\not{q}+m)}{(k+q)^2-m^2+\mathrm{i}\epsilon}\right]. \tag{2.322}$$

Gauge invariance implies that the result must be transverse:

$$q_\mu \Pi^{\mu\nu}(q) = 0. \tag{2.323}$$

Equivalently, the tensor must have the form

$$\Pi^{\mu\nu}(q) = \big(q^\mu q^\nu - q^2 g^{\mu\nu}\big)\Pi(q^2). \tag{2.324}$$

This transversality is not an optional simplification; it is the reason QED does not generate a photon mass counterterm $m_\gamma^2 A_\mu A^\mu$.

To see how the divergence appears, combine the two fermion denominators:

$$\frac{1}{(k^2-m^2)[(k+q)^2-m^2]} = \int_0^1 \mathrm{d}x\, \frac{1}{[(\ell^2-\Delta_\gamma(x))]^2}, \tag{2.325}$$

where

$$\ell = k + xq, \qquad \Delta_\gamma(x) = m^2 - x(1-x)q^2. \tag{2.326}$$

After the shift, terms odd in ℓ vanish. The tensor integrals are reduced by rotational symmetry:

$$\int \mathrm{d}^d\ell\, \ell^\mu \ell^\nu f(\ell^2) = \frac{g^{\mu\nu}}{d}\int \mathrm{d}^d\ell\, \ell^2 f(\ell^2). \tag{2.327}$$

Carrying out the Dirac trace and keeping only the ultraviolet pole gives

$$\Pi^{\mu\nu}_{\mathrm{div}}(q) = \frac{e^2}{12\pi^2}\frac{1}{\bar{\epsilon}}\big(q^\mu q^\nu - q^2 g^{\mu\nu}\big). \tag{2.328}$$

The counterterm from $-\frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu}$ contributes

$$\mathrm{i}\Pi^{\mu\nu}_{\mathrm{ct}}(q) = \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big). \tag{2.329}$$

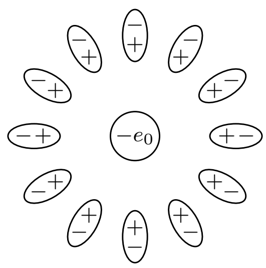
Thus the MS counterterm is

$$\delta_3^{\mathrm{MS}} = -\frac{e^2}{12\pi^2}\frac{1}{\epsilon}. \tag{2.330}$$

For N_f Dirac fermions with electric charges $Q_f e$, this generalizes to

$$\delta_3^{\mathrm{MS}} = -\frac{e^2}{12\pi^2}\frac{1}{\epsilon}\sum_{f=1}^{N_f} Q_f^2. \tag{2.331}$$

The same diagram also has a very direct physical meaning. The external photon creates a disturbance of the Dirac sea: for a short time, virtual electron-positron pairs are produced and then annihilated. In position space these pairs behave like tiny electric dipoles. Around a negative bare charge, the positive ends of the dipoles are pulled slightly inward and the negative ends are pushed slightly outward:



virtual e^+e^- dipoles screen the long-distance electric charge

This picture should not be taken as a literal classical cloud of particles. It is a useful way to remember the sign of the effect. The vacuum is polarized by the source charge. A probe very far away sees the source charge together with the induced dipoles, so the charge looks smaller. A probe at very short distance penetrates the polarization cloud and sees a larger effective charge.

The connection with the photon self-energy is as follows. In a static process the exchanged photon has $q^0 = 0$, so $q^2 = -\mathbf{q}^2$. The Dyson-resummed transverse photon propagator changes the Coulomb exchange from

$$\widetilde{V}_0(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2} \tag{2.332}$$

to

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2}\frac{1}{1-\Pi_R(-\mathbf{q}^2)}. \tag{2.333}$$

Here Π_R is the renormalized scalar vacuum-polarization function, and the on-shell normalization convention is

$$\Pi_R(0) = 0. \tag{2.334}$$

This condition means that α is the electric charge measured at very long distance, for example in the Thomson limit. Expanding the denominator to first order gives

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2}\big[1 + \Pi_R(-\mathbf{q}^2) + O(\alpha^2)\big]. \tag{2.335}$$

Fourier transforming this correction gives the Uehling correction to the Coulomb potential. For an electron source and a positive test charge,

$$V(r) = -\frac{\alpha}{r}[1 + \delta_{\mathrm{VP}}(r)], \tag{2.336}$$

where

$$\delta_{\mathrm{VP}}(r) = \frac{2\alpha}{3\pi}\int_1^\infty \mathrm{d}s\, \mathrm{e}^{-2mrs}\left(1 + \frac{1}{2s^2}\right)\frac{\sqrt{s^2-1}}{s^2}. \tag{2.337}$$

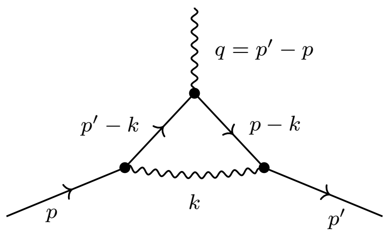
The factor e^{-2mr} reflects the threshold for making a virtual e^+e^- pair: the pair costs energy about $2m$. At distances much larger than the electron Compton wavelength, $r \gg m^{-1}$, the integral is dominated by $s \simeq 1$ and becomes

$$\delta_{\mathrm{VP}}(r) \simeq \frac{\alpha}{4\sqrt{\pi}}\frac{\mathrm{e}^{-2mr}}{(mr)^{3/2}}. \tag{2.338}$$

Thus the correction dies off exponentially at long distance. At shorter distances, or equivalently at larger momentum transfer $Q = \sqrt{-q^2}$, the probe resolves more of the unscreened charge. This is the physical origin of the running QED coupling.

3.5 Vertex correction and the Ward identity

The remaining superficially divergent one-loop graph is the correction to the electron-photon vertex:



Let the incoming electron momentum be p, the outgoing electron momentum be p', and the photon momentum be $q = p' - p$. We write the full renormalized vertex as

$$-\mathrm{i}e\mu^{\epsilon/2}\Gamma^\mu(p', p), \qquad \Gamma^\mu(p', p) = \gamma^\mu + \Lambda^\mu(p', p) + \delta_1\gamma^\mu. \tag{2.339}$$

The loop part is

$$\begin{aligned} -\mathrm{i}e\mu^{\epsilon/2}\Lambda^\mu(p', p) = (-\mathrm{i}e\mu^{\epsilon/2})^3 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\gamma^\alpha \frac{\mathrm{i}(\not{p}' - \not{k} + m)}{(p'-k)^2 - m^2 + \mathrm{i}\epsilon}\gamma^\mu \\ \times \frac{\mathrm{i}(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + \mathrm{i}\epsilon}\gamma^\beta \frac{-\mathrm{i}g_{\alpha\beta}}{k^2 + \mathrm{i}\epsilon}. \end{aligned} \tag{2.340}$$

The ultraviolet divergent part is local, so it cannot depend on the external momenta except through terms that are suppressed by powers of the loop momentum. Therefore the divergent part must be proportional to γ^μ:

$$\Lambda^\mu_{\mathrm{div}}(p', p) = \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}\gamma^\mu. \tag{2.341}$$

Finiteness of Γ^μ then gives

$$\delta_1^{\mathrm{MS}} = -\frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}. \tag{2.342}$$

This result is not independent of the electron self-energy. Gauge invariance implies the Ward–Takahashi identity

$$q_\mu \Gamma^\mu(p+q,p) = S^{-1}(p+q) - S^{-1}(p). \tag{2.343}$$

Taking the limit $q \to 0$ gives

$$\Gamma^\mu(p,p) = \frac{\partial S^{-1}(p)}{\partial p_\mu}. \tag{2.344}$$

Since

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}), \tag{2.345}$$

the loop correction obeys

$$\Lambda^\mu(p,p) = -\frac{\partial \Sigma(\not{p})}{\partial p_\mu}. \tag{2.346}$$

Using the divergent part of the self-energy,

$$\Sigma_{\rm div}(\not{p}) = \frac{e^2}{16\pi^2}\frac{1}{\bar\epsilon}\big(-\not{p} + 4m\big), \tag{2.347}$$

we find

$$-\frac{\partial \Sigma_{\rm div}}{\partial p_\mu} = \frac{e^2}{16\pi^2}\frac{1}{\bar\epsilon}\gamma^\mu, \tag{2.348}$$

which is exactly the vertex divergence above. Therefore

$$\delta_1^{\rm MS} = \delta_2^{\rm MS}, \qquad Z_1 = Z_2. \tag{2.349}$$

This equality is the practical content of the Ward identity in QED.

3.6 Renormalization constants and the running charge

Collecting the one-loop MS counterterms for one Dirac fermion of unit charge, we have

$$Z_2 = 1 - \frac{e^2}{16\pi^2}\frac{1}{\bar\epsilon} + O(e^4), \tag{2.350}$$

$$Z_m = 1 - \frac{3e^2}{16\pi^2}\frac{1}{\bar\epsilon} + O(e^4), \tag{2.351}$$

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\frac{1}{\bar\epsilon} + O(e^4), \tag{2.352}$$

$$Z_1 = Z_2 = 1 - \frac{e^2}{16\pi^2}\frac{1}{\bar\epsilon} + O(e^4). \tag{2.353}$$

Before extracting the running coupling, let us separate two ideas which are often compressed into one sentence. First, Z_3 is the photon field-strength renormalization:

$$A_0^\mu = Z_3^{1/2} A^\mu. \tag{2.354}$$

This means that the bare photon kinetic term becomes

$$\begin{aligned} -\frac{1}{4}F_{0\mu\nu}F_0^{\mu\nu} &= -\frac{1}{4}Z_3 F_{\mu\nu}F^{\mu\nu} \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu}. \end{aligned} \tag{2.355}$$

In momentum space the corresponding counterterm has the transverse form

$$\mathrm{i}\Pi^{\mu\nu}_{\mathrm{ct}}(q) = \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big). \tag{2.356}$$

This is exactly the tensor structure produced by the divergent part of the vacuum-polarization loop. Therefore the divergence changes the normalization of the photon kinetic term, not the photon mass. A photon mass term $m_\gamma^2 A_\mu A^\mu$ would not be transverse and is forbidden by gauge invariance.

Second, the electric charge is the coefficient of the interaction between the photon and the conserved electron current. Starting from the bare interaction,

$$\begin{aligned} -e_0\bar{\psi}_0\gamma^\mu\psi_0 A_{0\mu} &= -\mu^{\epsilon/2} e\, Z_e Z_2 Z_3^{1/2}\bar{\psi}\gamma^\mu\psi A_\mu \\ &= -\mu^{\epsilon/2} e\, Z_1\bar{\psi}\gamma^\mu\psi A_\mu, \end{aligned} \tag{2.357}$$

so that

$$Z_1 = Z_e Z_2 Z_3^{1/2}. \tag{2.358}$$

Equivalently,

$$e_0 = \mu^{\epsilon/2} Z_e e, \qquad Z_e = \frac{Z_1}{Z_2 Z_3^{1/2}}. \tag{2.359}$$

The Ward identity gives

$$Z_1 = Z_2. \tag{2.360}$$

Thus the vertex renormalization and the electron field-strength renormalization cancel out of the electric charge. The result is

$$Z_e = Z_3^{-1/2}. \tag{2.361}$$

This is the algebraic form of the physical statement that QED charge renormalization is caused by vacuum polarization. The virtual charged pairs modify the photon propagator, and this modification changes the electric charge inferred from photon exchange.

The same statement can be seen directly from a current-current scattering amplitude. If two conserved external currents exchange a photon, then only the transverse photon propagator matters:

$$\mathrm{i}\mathcal{M}(q) = [-\mathrm{i}eJ_\mu(q)]D^{\mu\nu}_{\mathrm{full}}(q)[-\mathrm{i}eJ'_\nu(-q)]. \tag{2.362}$$

Using

$$D^{\mu\nu}_{\mathrm{full}}(q) = \frac{-\mathrm{i}}{q^2[1-\Pi_R(q^2)]}\left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}\right) + \text{longitudinal part}, \tag{2.363}$$

and current conservation,

$$q_\mu J^\mu(q) = 0, \qquad q_\nu J'^\nu(-q) = 0, \tag{2.364}$$

the longitudinal part drops out. Thus one can package the effect of vacuum polarization into an effective charge,

$$e^2_{\mathrm{eff}}(q^2) = \frac{e^2}{1-\Pi_R(q^2)}. \tag{2.365}$$

For spacelike momentum transfer $q^2 = -Q^2$, this becomes

$$\alpha_{\mathrm{eff}}(Q) = \frac{\alpha}{1-\Pi_R(-Q^2)}. \tag{2.366}$$

At large Q^2 compared with the electron mass, the one-loop vacuum-polarization function contains

$$\Pi_R(-Q^2) = \frac{\alpha}{3\pi}\ln\frac{Q^2}{m^2} + \text{scheme-dependent finite terms} + O(\alpha^2). \tag{2.367}$$

Therefore the effective charge grows when the probe momentum Q grows:

$$\alpha_{\text{eff}}(Q) \simeq \frac{\alpha}{1 - \dfrac{\alpha}{3\pi}\ln\dfrac{Q^2}{m^2}}. \tag{2.368}$$

This is the same physics as the real-space screening picture: long distances see a screened charge, while short distances see more of the unscreened charge.

At one loop,

$$\begin{aligned} Z_e &= \left(1 - \frac{e^2}{12\pi^2}\frac{1}{\bar{\epsilon}}\right)^{-1/2} + O(e^4) \\ &= 1 + \frac{e^2}{24\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4). \end{aligned} \tag{2.369}$$

Since

$$\frac{1}{\bar{\epsilon}} = \frac{2}{\epsilon} - \gamma_E + \ln 4\pi, \tag{2.370}$$

the coefficient of the simple $1/\epsilon$ pole in Z_e is

$$Z_e = 1 + \frac{e^2}{12\pi^2}\frac{1}{\epsilon} + \text{finite MS convention terms} + O(e^4). \tag{2.371}$$

The bare charge does not depend on the arbitrary scale μ:

$$0 = \mu\frac{\mathrm{d}e_0}{\mathrm{d}\mu}. \tag{2.372}$$

Using

$$e_0 = \mu^{\epsilon/2} Z_e(e,\epsilon) e, \tag{2.373}$$

and extracting the coefficient of the simple pole in Z_e gives the one-loop beta function

$$\beta(e) \equiv \mu\frac{\mathrm{d}e}{\mathrm{d}\mu} = \frac{e^3}{12\pi^2} + O(e^5). \tag{2.374}$$

In terms of

$$\alpha = \frac{e^2}{4\pi}, \tag{2.375}$$

this is

$$\beta(\alpha) \equiv \mu\frac{\mathrm{d}\alpha}{\mathrm{d}\mu} = \frac{2}{3\pi}\alpha^2 + O(\alpha^3). \tag{2.376}$$

Integrating the one-loop equation between a reference scale μ and a momentum scale Q gives

$$\int_{\alpha(\mu)}^{\alpha(Q)} \frac{\mathrm{d}\alpha}{\alpha^2} = \frac{2}{3\pi}\int_{\mu}^{Q} \mathrm{d}\ln\mu', \tag{2.377}$$

$$\frac{1}{\alpha(Q)} = \frac{1}{\alpha(\mu)} - \frac{2}{3\pi}\ln\frac{Q}{\mu}. \tag{2.378}$$

Equivalently,

$$\alpha(Q) = \frac{\alpha(\mu)}{1 - \dfrac{2\alpha(\mu)}{3\pi}\ln\dfrac{Q}{\mu}} + O(\alpha^3). \tag{2.379}$$

Using $Q^2 = -q^2$ for spacelike momentum transfer, this is often written as

$$\alpha(Q) = \frac{\alpha(\mu)}{1 - \dfrac{\alpha(\mu)}{3\pi}\ln\dfrac{Q^2}{\mu^2}} + O(\alpha^3). \tag{2.380}$$

The positive sign of the QED beta function means that electric charge is screened in the infrared and becomes stronger in the ultraviolet. The one-loop formula would formally develop a Landau pole when the denominator vanishes, but that scale lies far outside the regime where perturbative QED alone should be trusted.

3.7 On-shell renormalization and physical interpretation

Minimal subtraction removes only ultraviolet poles. It is very convenient for deriving renormalization-group equations, but it does not define the parameters directly by physical measurements. In the on-shell scheme the renormalized mass is the pole of the electron propagator, the residue of that pole is one, and the electric charge is measured in the Thomson limit. These conditions are

$$S_R^{-1}(p)\Big|_{\not p = m} = 0, \tag{2.381}$$

$$\frac{\partial S_R^{-1}(p)}{\partial \not p}\bigg|_{\not p = m} = 1, \tag{2.382}$$

$$\Pi_R(q^2 = 0) = 0, \tag{2.383}$$

$$\Gamma_R^\mu(p,p)\Big|_{\not p = m} = \gamma^\mu. \tag{2.384}$$

The first two equations fix the mass and wave-function counterterms of the electron. The third equation fixes the normalization of the photon field so that the photon remains massless and the long-distance Coulomb law has the observed normalization. The last equation fixes the electric charge.

There is one extra subtlety in QED: the photon is massless, so on-shell Green functions can also contain infrared divergences. These are not UV renormalization divergences. They cancel in inclusive physical observables once real soft-photon emission is included. The UV counterterms derived above are local and are independent of that infrared physics.

The final picture is therefore simple. QED needs exactly four local renormalizations:

$$\psi \text{ field strength}, \qquad m \text{ mass}, \qquad A_\mu \text{ field strength}, \qquad e \text{ charge}. \tag{2.385}$$

The Ward identity reduces the number of independent charge-related renormalizations by enforcing $Z_1 = Z_2$, so the running of e is governed by Z_3, namely by vacuum polarization.

4. Wilson's approach to renormalization

4.1 Cut-off in high energy scale

The main idea of Wilson's renormalization is introducing a cut-off Λ and compute how the theory changes at different length scales. For that, we need to working with the generating functional with cut-off and successive integration over large momenta/small length scales:

$$Z = \int \mathcal{D}\phi_\Lambda \, e^{-S[\phi]}, \tag{2.386}$$

where we used Wick rotation changing to Euclidean space-time:

$$x^0 \to \mathrm{i}x^4, \quad \mathrm{d}^4x \to \mathrm{i}\mathrm{d}^4x_E. \tag{2.387}$$

Thus, the measurement is

$$\mathcal{D}\phi_\Lambda = \prod_{|k|<\Lambda} \mathrm{d}\phi(k). \tag{2.388}$$

In this case, the modes of ϕ are well defined in Euclidean space-time such that we can make sure that when $k \to \infty$ all the components of momentum also go to infinity. This step is to avoid in Minkowski space-time $k^2 = (k^0)^2 - (k^i)^2$. So when $k^2 \to \infty$, it could be $k^0 \to \infty$ and $k^i \to 0$, which doesn't suitable for our analysis.

Now, the action and Lagrangian become

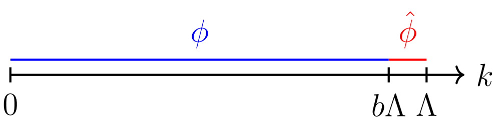
$$S[\phi] = \int \mathrm{d}^4x \mathcal{L}, \tag{2.389}$$

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4. \tag{2.390}$$

Suppose the mass $m \ll \Lambda$, and the coupling constant $\lambda \ll 1$, which means the last two terms in the Lagrangian can be treated as perturbation. For going to the perturbation theory, it's useful to split up fields $\phi \to \phi + \hat{\phi}$ as

$$\hat{\phi} = \begin{cases} \phi(k) & \text{for } b\Lambda \le |k| \le \Lambda \\ 0 & \text{for } |k| \le b\Lambda \end{cases}; \qquad \phi = \begin{cases} 0 & \text{for } b\Lambda \le |k| \le \Lambda \\ \phi(k) & \text{for } |k| \le b\Lambda \end{cases}, \tag{2.391}$$

where $(b\Lambda, \Lambda)$ is a very small interval controlled by parameter $b \le 1$



Then we can also split up the action as $S[\phi] \to S[\phi] + \hat{S}[\phi, \hat{\phi}]$, so we have

$$\begin{aligned} S[\phi] &= \int \mathrm{d}^4x \frac{1}{2}\partial_\mu\left(\phi + \hat{\phi}\right)\partial^\mu\left(\phi + \hat{\phi}\right) + \frac{1}{2}m^2\left(\phi + \hat{\phi}\right)^2 + \frac{\lambda}{4!}\left(\phi + \hat{\phi}\right)^4 & (2.392) \\ &= \int \mathrm{d}^4x \frac{1}{2}\left(\partial_\mu\phi\partial^\mu\phi + 2\partial_\mu\phi\partial^\mu\hat{\phi} + \partial_\mu\hat{\phi}\partial^\mu\hat{\phi}\right) + \frac{1}{2}m^2\left(\phi^2 + \hat{\phi}^2 + 2\phi\hat{\phi}\right) & (2.393) \\ &\qquad + \frac{\lambda}{4!}\left(\phi^4 + \phi^2\hat{\phi}^2 + 2\phi^3\hat{\phi} + \hat{\phi}^2\phi^2 + \hat{\phi}^4 + 2\phi\hat{\phi}^3 + 2\phi^3\hat{\phi} + 2\phi\hat{\phi}^3 + 4\phi^2\hat{\phi}^2\right) & (2.394) \\ &= \int \mathrm{d}^4x \left[\underbrace{\frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4}_{\mathcal{L}[\phi]} + \underbrace{\frac{1}{2}\left(\partial_\mu\hat{\phi}\right)^2 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{4!}\left(6\phi^2\hat{\phi}^2 + 4\phi^3\hat{\phi} + 4\phi\hat{\phi}^3 + \hat{\phi}^4\right)}_{\hat{\mathcal{L}}[\phi,\hat{\phi}]}\right], & (2.395) \end{aligned}$$

where we have dropped terms containing the products $\phi\hat{\phi}$ because Fourier components are orthogonal:

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{d}^4k'}{(2\pi)^4}\phi(k)\hat{\phi}(k')e^{\mathrm{i}(k+k')\cdot x} \sim \delta^{(4)}(k+k') = 0 \tag{2.396}$$

Thus, we have

$$\hat{S}[\phi,\hat{\phi}] = \int \mathrm{d}^4x(\hat{\mathcal{L}}_0[\hat{\phi}] + \hat{\mathcal{L}}_{\mathrm{int}}[\phi,\hat{\phi}]), \tag{2.397}$$

and

$$\hat{\mathcal{L}}_0[\hat{\phi}] = \frac{1}{2}\partial_\mu\hat{\phi}\partial^\mu\hat{\phi}, \tag{2.398}$$

$$\hat{\mathcal{L}}_{\mathrm{int}}[\phi,\hat{\phi}] = \frac{1}{2}m^2\hat{\phi}^2 + \lambda\left(\frac{1}{3!}\phi\hat{\phi}^3 + \frac{1}{4}\phi^2\hat{\phi}^2 + \frac{1}{3!}\phi^3\hat{\phi} + \frac{1}{4!}\hat{\phi}^4\right). \tag{2.399}$$

Thus, the first part in $\hat{S}$ can be rewritten by the Fourier transformation as

$$\int \mathrm{d}^4x\, \hat{\mathcal{L}}_0[\hat{\phi}] = \int \frac{\mathrm{d}^4k\mathrm{d}^4p}{(2\pi)^8}\left(\mathrm{i}k_\mu\hat{\phi}(k)\right)\left(\mathrm{i}p^\mu\hat{\phi}(p)\right)\int \mathrm{d}^4x\, e^{\mathrm{i}(k+p)\cdot x} \tag{2.400}$$

$$= \int \frac{\mathrm{d}^4k\mathrm{d}^4p}{(2\pi)^8}(\mathrm{i}k_\mu)(\mathrm{i}p^\mu)\hat{\phi}(k)\hat{\phi}(p)\,(2\pi)^4\delta^{(4)}\,(k+p) \tag{2.401}$$

$$= \frac{1}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\hat{\phi}(-k)k^2\hat{\phi}(k) \tag{2.402}$$

$$= \frac{1}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\hat{\phi}^*(k)k^2\hat{\phi}(k). \tag{2.403}$$

This Gaussian integral form remind us to write a propagator for $\hat{\phi}$ from Wick contraction of $\hat{\phi}$ fields:

$$\hat{\phi}(k)\hat{\phi}(p) = \langle\hat{\phi}(k)\hat{\phi}(p)\rangle = \frac{\int \mathcal{D}\hat{\phi}\,\hat{\phi}(k)\hat{\phi}(p)e^{-S_0[\hat{\phi}]}}{\int \mathcal{D}\hat{\phi}e^{-S_0[\hat{\phi}]}} = \frac{1}{k^2}(2\pi)^4\delta^{(4)}(k+p)\Theta(k), \tag{2.404}$$

where

$$\Theta(k) = \begin{cases} 1 & \text{for } b\Lambda < |k| < \Lambda \\ 0 & \text{else}\end{cases}. \tag{2.405}$$

4.2 Effective action and ideas about renormalization group

Therefore, we can separate the generating functional measurement with the **high energy modes** $\hat{\phi}$ and **low energy modes** $\phi_{b\Lambda}$ as $\mathcal{D}\phi_\Lambda = \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda$. And we can integrate the high energy modes

$$Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]} = \int \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda e^{-S[\phi]-\hat{S}[\phi,\hat{\phi}]} = \int \mathcal{D}\phi_{b\Lambda}e^{-S[\phi]}\left(\int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]}\right) \tag{2.406}$$

$$= \int \mathcal{D}\phi_{b\Lambda}e^{-S_{\mathrm{eff}}[\phi]}, \tag{2.407}$$

where left a **effective action** defined as

$$S_{\mathrm{eff}}[\phi] = S[\phi] + \delta S[\phi], \tag{2.408}$$

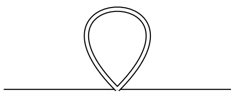
in which $\delta S[\phi]$ can be treated as perturbation, thus it can be expanded as

$$
\begin{aligned}
e^{-\delta S[\phi]} &= \int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\int \mathrm{d}^4x \hat{\mathcal{L}}[\phi,\hat{\phi}]} && (2.409)\\
&\simeq \int \mathcal{D}\hat{\phi}_\Lambda \int \mathrm{d}^4x \left[1 - \left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi\hat{\phi}^3 + \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi^3\hat{\phi} + \frac{\lambda}{4!}\hat{\phi}^4\right) + \mathcal{O}(\lambda^2)\right] && (2.410)\\
&= \int \mathcal{D}\hat{\phi}_\Lambda \int \mathrm{d}^4x \Bigg[1 - \underbrace{\left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{4!}\hat{\phi}^4\right)}_{\substack{\text{Doesn't contain } \phi,\\ \text{which has no contribution for running}}} - \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \mathcal{O}(\lambda^2)\Bigg], && (2.411)
\end{aligned}
$$

where in the third step we have dropped all the odd terms which don't contribute to the integral. The introducing of $\delta S[\phi]$ will change the parameters in the original action $S[\phi]$, which will finally leading the parameters start running. To show this, we now need to work in the perturbation scenario. Since we have assumed that $m \ll \Lambda$ and $\lambda \ll 1$, so we can expand $e^{-S_{\text{eff}}[\phi]}$ with Dyson series in terms of λ. For that, we can generally write the effective Lagrangian $\mathcal{L}_{\text{eff}}$ for effective action $S_{\text{eff}}[\phi]$ as:

$$
\mathcal{L}_{\text{eff}}[\phi] = \frac{1}{2}(1+\Delta Z)\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}(m^2+\Delta m^2)\phi^2 + \frac{1}{4!}(\lambda+\Delta\lambda)\phi^4 + (\text{higher dimensional operators}). \tag{2.412}
$$

Now the mission is to determine ΔZ, Δm^2 and $\Delta\lambda$. We firstly consider the 1-loop order $\mathcal{O}(\lambda)$ correction for the propagator, which corresponds to the term $\frac{1}{4}\lambda\phi^2\hat{\phi}^2$, the graph is :

$I_1 = \phi$  ϕ , (2.413)

$\mathcal{O}(\lambda)$

where the double line represent the contraction of $\hat{\phi}$, which doesn't contribute the real physical observable, thus it needs to be integrated. Thus, we have

$$
\begin{aligned}
I_1 &= -\frac{\lambda}{4}\int \mathrm{d}^4x \phi^2 \overbracket{\hat{\phi}\hat{\phi}} && (2.414)\\
&= -\frac{\lambda}{4}\int \mathrm{d}^4x \int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_4}{(2\pi)^4} e^{\mathrm{i}(k_1+\cdots+k_4)\cdot x}\phi(k_1)\phi(k_2)\frac{1}{k_3^2}(2\pi)^4\delta^{(4)}(k_3+k_4)\Theta(k_3), && (2.415)
\end{aligned}
$$

integrate for x, k_4:

$$
\begin{aligned}
I_1 &= -\frac{\lambda}{4}\int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_3}{(2\pi)^4}\phi(k_1)\phi(k_2)\frac{1}{k_3^2}(2\pi)^4\delta^{(4)}(k_1+k_2)\Theta(k_3) && (2.416)\\
&= -\frac{1}{2}\mu\int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\phi(k_1)\phi(-k_1), && (2.417)
\end{aligned}
$$

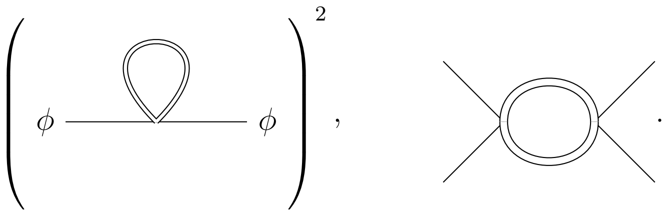
where

$$
\begin{aligned}
\mu &= \frac{\lambda}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\Theta(k)}{k^2} && (2.418)\\
&= \frac{\lambda}{2}\frac{1}{(2\pi)^4}\int \mathrm{d}\Omega \int_{b\Lambda}^{\Lambda} k\mathrm{d}k && (2.419)\\
&= \frac{\lambda}{32\pi^2}(1-b^2)\Lambda^2. && (2.420)
\end{aligned}
$$

I_1 has structure of mass correction, which can be put in exponent of effective action as

$$\exp\left\{-\int \mathrm{d}^4x \frac{1}{2}\mu\phi^2\right\}. \tag{2.421}$$

And the $\mathcal{O}(\lambda^2)$ order graphs therefore are:



For the first one, it's nothing but the disconnected graph $\sim \frac{1}{2}I_1^2$, which contribute $\mathcal{O}(\lambda^2)$ in the exponent. The second graph represent the correction for the ϕ^4 interaction vertex:

$$I_2 = -\frac{1}{4!}\xi \int \mathrm{d}^4x\phi^4(x), \tag{2.422}$$

factorizing the loop integral, then we have

$$\xi = -4! \cdot \underbrace{2}_{\substack{\text{2 possible}\\ \text{contractions}\\ \text{of } \phi^2(\hat{\phi}\hat{\phi})^2}} \cdot \underbrace{\frac{1}{2!}}_{\substack{\text{symmetry factor}\\ \text{of diagram}}} \cdot \left(\frac{\lambda}{4}\right)^2 \int_{b\Lambda<|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\left(\frac{1}{k^2}\right)^2\Theta(k) = -\frac{3\lambda^2}{16\pi^2}\ln\frac{1}{b} \tag{2.423}$$

Thus, we find μ, ξ can be interpreted as combinations to $\Delta m^2, \Delta\lambda$.

Therefore, the generating functional $Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}$ changing to $Z = \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}$ is by rescaling S to effective action S_{eff} by changing the coordinates and momenta:

$$k' = \frac{k}{b} \quad , \quad x' = bx \tag{2.424}$$

Thus,

$$S_{\text{eff}} = \int \mathrm{d}^4x' \frac{1}{b^4}\Bigg(\frac{1}{2}(1+\Delta Z)b^2\partial'_\mu\phi\partial'^\mu\phi + \frac{1}{2}(m^2+\Delta m^2)\phi^2 + \frac{1}{4!}(\lambda+\Delta\lambda)\phi^4 + \underbrace{\cdots + \frac{1}{n!}(\lambda_n+\Delta\lambda_n)\phi^n + \dots}_{\text{higher dim. operators}}\Bigg) \tag{2.425}$$

$$= \int \mathrm{d}^4x' \left(\frac{1}{2}\partial'_\mu\phi'\partial'^\mu\phi' + \frac{1}{2}m'^2\phi'^2 + \frac{1}{4!}\lambda'\phi'^4 + \cdots + \frac{1}{n!}\lambda'_n(\phi')^n + \dots\right), \tag{2.426}$$

where

$$\phi' = \sqrt{(1+\Delta Z)b^{-2}}\phi \tag{2.427}$$
$$m'^2 = (m^2+\Delta m^2)(1+\Delta Z)^{-1}b^{-2} \tag{2.428}$$
$$\lambda' = (\lambda+\Delta\lambda)(1+\Delta Z)^{-2}b^0 \tag{2.429}$$
$$\lambda'_n = (\lambda_n+\Delta\lambda_n)(1+\Delta Z)^{-n/2}b^{n-4}. \tag{2.430}$$

This process, mapping $\phi' \to \phi$, $m'^2 \to m^2$, $\lambda' \to \lambda$, is exactly the **renormalization group transformations**. We can also explicitly show the relations for rescaling of parameters for mapping $x' = bx$ on $D = 4$ dimension:

$$m'^2 = b^{-2}\Delta m^2 \tag{2.431}$$
$$\lambda' = b^0\Delta\lambda \tag{2.432}$$
$$\lambda'_n = b^{n-4}\Delta\lambda_n \tag{2.433}$$

Generally, in D dimensions, we have:

$$m'^2 = b^{-2}\Delta m^2 \tag{2.434}$$
$$\lambda' = b^{D-4}\Delta\lambda \tag{2.435}$$
$$\lambda'_n = b^{D-4+n-4}\Delta\lambda_n \tag{2.436}$$

for $b < 1$ on $D = 4$:

- Super-renormalizable m^2 grows as $b^{-2} \longleftrightarrow$ relevant operator ϕ^2.
- Renormalizable λ scales as $b^0 \longleftrightarrow$ marginal operators $\lambda\phi^4$.
- Non-renormalizable λ_n vanishes as $b^{n-4} \longleftrightarrow$ irrelevant operators $\lambda_n\phi^n, n > 4$.

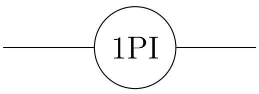
$$\lambda' = \lambda - \frac{3}{16\pi^2}\lambda^2 \ln\left(\frac{1}{b}\right) + O(\lambda^3) \tag{2.437}$$

5. Callan-Symanzik Equation

Goal: Quantify scale dependence of Green's functions $G^{(n)}(x_1 \ldots x_n, M)$, where M sets the scale of external momenta, which leads to the Callan-Symanzik equation.

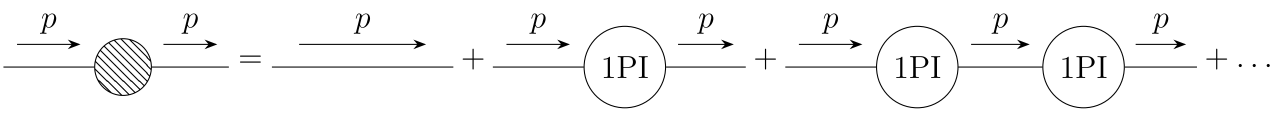
5.1 Alternative choice for renormalization in ϕ^4-theory

In ϕ^4-theory, the 1-PI graphs of propagator can be renormalized by taking the remormalization scheme. We simply recall this process again, firstly, taking



$$= -\mathrm{i}\Sigma(p^2), \tag{2.438}$$

for all 1-PI contributions to propagators. Thus, the full propagator goes like:

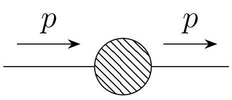


$$= \frac{\mathrm{i}}{p^2 - m^2 - \Sigma(p^2)}. \tag{2.439}$$

Here we took the on-shell scheme:

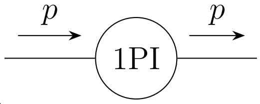
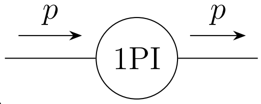
$$\Sigma(p^2)\big|_{p^2=m_{ph}^2} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}p^2}\Sigma(p^2)\big|_{p^2=m_{ph}^2} = 0. \tag{2.440}$$

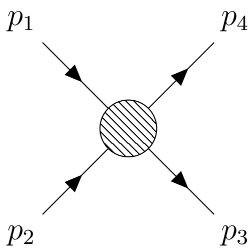
This scheme force the residue around the pole to be unity. Then we obtained the renormalized propagator:



$$= \frac{\mathrm{i}}{p^2 - m_{ph}^2}. \tag{2.441}$$

But this is not physical in many cases, say QCD and SM due to many of problems. For example, since the quarks are confined so normally we can hardly define the asymptotic state to imagine a free quark. And the massless field of SM in this scheme will meet IR-divergence, which will also lead us to many unnecessary problems. So we need to choose other better schemes for a more general model. One alternative is to define counterterms at arbitrary scale M, at the external space-like momenta $p^2 = -M^2 < 0$, we take

$\Bigg($  $\Bigg)_{p^2=-M^2} = 0 \quad , \quad \frac{\mathrm{d}}{\mathrm{d}p^2}\Bigg($  $\Bigg)_{p^2=-M^2} = 0,$

$\Bigg($  $\Bigg)_{s=t=u=-M^2} = \mathrm{i}\lambda. \qquad (2.442)$

where $(p_1+p_2)^2 = s$, $(p_1+p_3)^2 = t$, $(p_1+p_4)^2 = u$. The reason why we choose the space-like momentum is to avoid poles. If we choose a time-like momentum, it for sure corresponds to physical particle, however also having divergence. The first condition $\Sigma(-M^2) = 0$ requires the correction for mass Σ_{finite} is 0 at this non-fixing scale M, and all the divergence Σ_{div} will be subtracted by the counterterms. So basically it looks very similar with the on-shell condition, the only thing that we change is changing the fixed physical mass m^2 to the unphysical mass scale $-M^2$.

This arbitrary parameter M is called the **renormalization scaling**. These conditions defined the only certain vale at a fixed point, which canceled all the divergent. So we say **it defines the theory in scale $\boldsymbol{M}$**.

For seeing this scheme is indeed valid, we firstly look at the Green's function with renormalized fields $\phi = \sqrt{Z}\phi_r$ and coupling λ, and taking the renormalized scale as M, we have

$$\begin{aligned} G^{(n)}_{\text{ren}}(x_1 \dots x_n, \lambda, M) &= \langle \Omega | T\{\phi_r(x_1)\dots\phi_r(x_n)\}|\Omega\rangle \\ &= Z^{-n/2}\langle \Omega | T\{\phi(x_1)\dots\phi(x_n)\}|\Omega\rangle \\ &= Z^{-n/2} \underbrace{G^{(n)}_{\text{bare}}(x_1\dots x_n, \lambda, M)}_{\text{bare Green's fun. independent of } M} . \end{aligned} \tag{2.443}$$

Now we change variables as $M \to M + \delta M$. This changing of our scaling will also change the other parameters to satisfy the renormalization condition. Thus it leads to $\lambda \to \lambda + \delta\lambda$ and $\phi_{\text{ren}} \to \phi'_{\text{ren}} = (1+\delta\eta)\phi_{\text{ren}}$, thus

$$(Z')^{-1/2}\phi_0 = Z^{-1/2}(1+\delta\eta)\phi_0 \implies Z' = Z(1+\delta\eta)^{-2} \simeq Z(1-2\delta\eta) \equiv Z(1+\delta Z), \tag{2.444}$$

where $\delta\eta = \frac{-\delta Z}{2}$. The latter gives the way of Green's function changing:

$$G^{(n)}_{\text{ren}} \to Z^{-n/2}(1+\delta Z)^{-n/2} G^{(n)}_{\text{bare}} \tag{2.445}$$

$$\simeq Z^{-n/2}(1 - \frac{n}{2}\delta Z) G^{(n)}_{\text{bare}} \tag{2.446}$$

$$= Z^{-n/2}(1 + n\delta\eta) G^{(n)}_{\text{bare}} \tag{2.447}$$

$$= (1+n\delta\eta) G^{(n)}_{\text{ren}}, \tag{2.448}$$

We will ignore the corner index "ren" in $G^{(n)}_{\text{ren}}$, **namely the renormalized Green's function denotes as $\boldsymbol{G^{(n)}}$.** Thus from $\delta G_{\text{bare}} = 0$, we have

$$0 = \delta(Z^{n/2}G^{(n)}) = \underbrace{(\delta Z^{n/2})G^{(n)}}_{n\delta\eta Z^{n/2}} + Z^{n/2}\frac{\partial}{\partial M}G^{(n)}\delta M + Z^{n/2}\frac{\partial}{\partial\lambda}G^{(n)}\delta\lambda. \tag{2.449}$$

Introducing the new variables:

$$\beta = \frac{M}{\delta M}\delta\lambda, \quad \gamma = \frac{M}{\delta M}\delta\eta \tag{2.450}$$

Thus, dropping the arbitrary δM, we obtain **the Callan-Symanzik equation**:

$$\left(M\frac{\partial}{\partial M} + \beta\frac{\partial}{\partial\lambda} + n\gamma\right) G^{(n)}(x_1 \dots x_n, \lambda, M) = 0, \tag{2.451}$$

where β and γ are universal for a given theory, which is independent for particular Green's function. This also means that these parameters are independent of any UV-cut off (dimensionless).

$$\beta = \beta(\lambda), \qquad \gamma = \gamma(\lambda). \tag{2.452}$$

We can also directly compute $\mathrm{d}G^{(n)}$ as

$$\mathrm{d}G^{(n)} = (1 + n\delta\eta)G^{(n)} - G^{(n)} = n\delta\eta G^{(n)} = n\frac{\mathrm{d}M}{M}\left(M\frac{\mathrm{d}\eta}{\mathrm{d}M}\right)G^{(n)} \tag{2.453}$$

$$= \frac{\partial G^{(n)}}{\partial M}\mathrm{d}M + \frac{\partial G^{(n)}}{\partial\lambda}\mathrm{d}\lambda \tag{2.454}$$

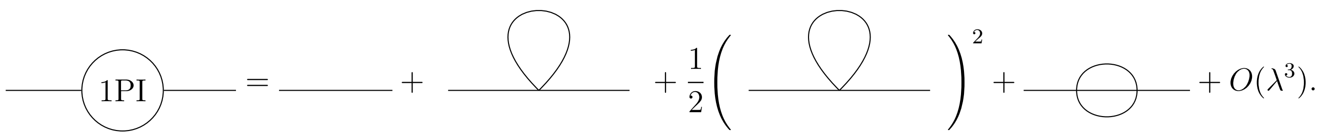
$$= \frac{\partial G^{(n)}}{\partial M}\mathrm{d}M + \left(M\frac{\mathrm{d}\lambda}{\mathrm{d}M}\right)\frac{\mathrm{d}M}{M}\frac{\partial G^{(n)}}{\partial\lambda} \tag{2.455}$$

$$= \frac{\mathrm{d}M}{M}\left[M\frac{\partial}{\partial M} + \left(M\frac{\mathrm{d}\lambda}{\mathrm{d}M}\right)\frac{\partial}{\partial\lambda}\right]G^{(n)}, \tag{2.456}$$

from this construction, we can also obtain the Callan-Symanzik equation.

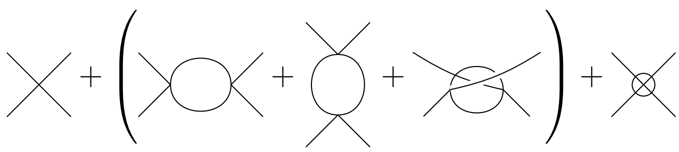
5.2 β-function in massless ϕ^4-theory

Now let's focus some actual cases to find the β-function and the anomalous dimension $\gamma(\lambda)$. For simplicity, we consider the 1-loop correction for **massless ϕ^4-theory**. The Callan-Symanzik equation for $G^{(2)}$ gives us $\gamma = 0 + O(\lambda^2)$, because in 1-loop order, there's no contribution to Z, but only to δm^2. Only until 2-loops or higher orders which contribute Z and also of course δm^2.



(2.457)

For $\beta(\lambda)$ at $O(\lambda)$, we need to consider the 4-point function correction:

$G^{(4)}(p) =$  $+ O(\lambda^3)$ (2.458)

$$= -\mathrm{i}\lambda + (-\mathrm{i}\lambda)^2(\mathrm{i}V(t) + \mathrm{i}V(s) + \mathrm{i}V(u)) - \mathrm{i}\delta\lambda + O(\lambda^3), \tag{2.459}$$

where the renormalization condition are taken in $t = s = u = -M^2$. So for each diagram vertex correction, for $m = 0$ we have:

$$V(p^2) = \frac{1}{2}\int\frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2}\frac{\mathrm{i}}{(p-k)^2}. \tag{2.460}$$

Counterterm $\delta\lambda$ determined from amputated Green's function

$$G^{(4)}_{\text{amp}}\big|_{s=t=u=-M^2} \to \delta\lambda = (-\mathrm{i}\lambda)^2 3V(-M^2). \tag{2.461}$$

Take $D = 4 - 2\epsilon$:

$$V(-M^2) = \lim_{\epsilon\to 0}\frac{1}{2}\frac{-1}{(4\pi)^{d/2}}\int_0^1 \mathrm{d}x\,\frac{\Gamma(\epsilon)}{(x(1-x)M^2)^\epsilon} \tag{2.462}$$

$$= -\frac{1}{32\pi^2}\left(\frac{1}{\epsilon} - \ln M^2 + O(1)\right). \tag{2.463}$$

Thus

$$\delta\lambda = \frac{3\lambda^2}{32\pi^2}\left(\frac{1}{\epsilon} - \ln M^2 + O(1)\right). \tag{2.464}$$

This will lead to

$$G^{(4)} = -\mathrm{i}\lambda + 3\mathrm{i}(\mathrm{i}\lambda)^2 V(p^2) - \mathrm{i}\delta\lambda + \mathcal{O}(\lambda^3) \tag{2.465}$$

$$= -\mathrm{i}\lambda + \frac{3\mathrm{i}\lambda^2}{32\pi^2}\left(\frac{1}{\epsilon} - \ln p^2 - \frac{1}{\epsilon} + \ln M^2\right) + \mathcal{O}(\lambda^3) \tag{2.466}$$

$$= -\mathrm{i}\lambda + \frac{3\mathrm{i}\lambda^2}{32\pi^2}\ln\left(\frac{M^2}{p^2}\right) + \mathcal{O}(\lambda^3) \tag{2.467}$$

Thus, we have

$$M\frac{\partial}{\partial M}G^{(4)} = \frac{3\mathrm{i}\lambda^2}{16\pi^2} + O(\lambda^3). \tag{2.468}$$

Thus, bring it into the Callan-Symanzik equation, we have

$$-\beta(\lambda)\frac{\partial}{\partial\lambda}G^{(4)} = \mathrm{i}\frac{3\lambda^2}{16\pi^2} + O(\lambda^3) \tag{2.469}$$

where $\gamma = 0 + O(\lambda^2)$. Since we only consider order $\mathcal{O}(\lambda^2)$, therefore we only need to expand $G^{(4)}$ into $\mathcal{O}(\lambda)$, i.e.,

$$\frac{\partial}{\partial\lambda}G^{(4)} = -\mathrm{i} + \frac{3\mathrm{i}\lambda}{16\pi^2}\ln\left(\frac{M^2}{p^2}\right) + \mathcal{O}(\lambda^2) \simeq -\mathrm{i} + \mathcal{O}(\lambda), \tag{2.470}$$

thus the β-function at 1-loop is:

$$\beta(\lambda) = \frac{3}{16\pi^2}\lambda^2 + O(\lambda^3). \tag{2.471}$$

5.3 Analogous analysis for massless QED

Now we consider the QED, which as a good example to calculate the anomalous dimension $\gamma(\lambda)$. In QED, we need to renormalize the vertex(Z_1) and the propagator for electron(Z_2) and photon(Z_3). These three parameters defined the electron charge renormalization:

$$e_0 = Z_1^{-1} Z_2 Z_3^{1/2} e. \tag{2.472}$$

The Green's function $G^{(n_e,n_A)}(x_1\ldots x_n, e, M)$ now contains two fields, so the Callan-Symanzik equation is

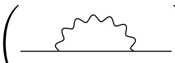
$$\left(M\frac{\partial}{\partial M} + \beta(e)\frac{\partial}{\partial e} + n_e\gamma_2 + n_A\gamma_3\right)G^{(n_e,n_A)}(x_1\ldots x_n, e, M) = 0, \tag{2.473}$$

where n_e denote for the electron fields, n_A for photon fields. And γ_2, γ_3 are anomalous dimensions of electron and photon field respectively

$$\gamma_i = \frac{1}{2} M \frac{\partial}{\partial M} \delta_i, \quad i = 2, 3. \tag{2.474}$$

The γ_i comes from the field renormalization, which related with Z_2 and Z_3. And $Z_i = 1 + \delta_i$, where δ_i is the counterterms of fields. So for $i = 2$, i.e., the electron field

$$\delta_2 = \frac{\mathrm{d}}{\mathrm{d}\not{p}} \Sigma(p)\Big|_{p^2=-M^2} = \frac{\mathrm{d}}{\mathrm{d}\not{p}} \left(\quad \right)$$



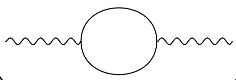
$$= \frac{-e^2}{16\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E \right) (1 - \epsilon \ln M)$$
$$= \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right), \tag{2.475}$$

which leads to

$$\gamma_2 = \frac{e^2}{16\pi^2} \equiv \frac{\alpha}{4\pi}. \tag{2.476}$$

Also $i = 3$ is for the photon field

$$\delta_3 = \Pi(p^2)\Big|_{p^2=-M^2} = \left(\quad \right) = \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \Pi(p^2)$$



$$= \frac{-e^2}{12\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{12\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right). \tag{2.477}$$

Thus,

$$\gamma_3 = \frac{e^2}{12\pi^2} \equiv \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \tag{2.478}$$

For calculating the β-function, we first briefly review a very important solution in QED renormalization. Consider the Ward-Takahashi identity

$$(p - q)_\mu \Gamma^\mu(p, q) = S_{\mathrm{F}}^{-1}(p) - S_{\mathrm{F}}^{-1}(q), \tag{2.479}$$

when we take the limit that $p \to q$ and move it to the right, thus we have

$$\Gamma^\mu(p, p) = \frac{\partial S_{\mathrm{F}}^{-1}(p)}{\partial p_\mu}. \tag{2.480}$$

Notice $S_{\mathrm{F}}^{-1}(p) = \gamma^\mu p_\mu - m - \Sigma(p)$, thus we have

$$\Gamma^\mu(p, p) = \gamma^\mu - \frac{\partial \Sigma(p)}{\partial p_\mu}. \tag{2.481}$$

Beside the momentum p, the propagator can be renormalized as

$$S_{\mathrm{F}}^{-1}(p) = Z_2^{-1}(\gamma \cdot p - m_{\mathrm{ph}}) + \ldots, \tag{2.482}$$

which means

$$\frac{\partial S_{\mathrm{F}}^{-1}}{\partial p_\mu} \approx Z_2^{-1} \gamma^\mu. \tag{2.483}$$

And for the renormalized vertex, we have

$$\Gamma^\mu = Z_1^{-1}\Gamma_R^\mu. \tag{2.484}$$

In low energy approximation, the vertex just $\Gamma_R^\mu \to \gamma^\mu$. Thus, comparing (2.481) and (2.484), we have

$$Z_1 = Z_2. \tag{2.485}$$

Thus, $\delta_1 = \delta_2$. Then we immediately have

$$\begin{aligned}
0 = \frac{\partial e_0}{\partial M} &= \frac{\partial}{\partial M}\left[e\left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)\right] && (2.486)\\
&= e\frac{\partial}{\partial M}\left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)+\left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)\frac{\partial e}{\partial M} && (2.487)\\
&\simeq e\frac{\partial}{\partial M}\left(-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)+\frac{\partial e}{\partial M}. && (2.488)
\end{aligned}$$

Therefore, we get the final result for the β-function:

$$\begin{aligned}
\beta(e) = M\frac{\partial}{\partial M}e &= -eM\frac{\partial}{\partial M}\left(\underbrace{-\delta_1+\delta_2}_{\delta_1=\delta_2}+\frac{1}{2}\delta_3\right)\\
&= \frac{e^3}{12\pi^2} \equiv e\frac{\alpha}{4\pi}\left(\frac{4}{3}\right). && (2.489)
\end{aligned}$$

6. Evolution of coupling

6.1 Solution of Callan-Symanzik equation

Now we would like to solve the solutions of Callan-Symanzik equation. We firstly start with the 2-point function $G^{(2)}(p)$, which generally can be written as:

$$G^{(2)}(p) = \frac{\mathrm{i}}{p^2} f\left(\frac{-p^2}{M^2}\right), \tag{2.490}$$

where $f\left(\frac{-p^2}{M^2}\right)$ is a dimensionless function, which only depends on $\frac{-p^2}{M^2}$. In momentum space, without the amputation, which means we add the external lines to contribute the momentum p in the propagator. We have:

$$\begin{aligned}
M\frac{\partial}{\partial M}G^{(2)}(p) &= \frac{\mathrm{i}}{p^2}\left(\frac{2p^2}{M^2}\right)f'\left(\frac{-p^2}{M^2}\right), && (2.491)\\
p\frac{\partial}{\partial p}G^{(2)}(p) &= -2G^{(2)}(p) - \frac{\mathrm{i}}{p^2}\left(\frac{2p^2}{M^2}\right)f'\left(\frac{-p^2}{M^2}\right). && (2.492)
\end{aligned}$$

Therefore, we have:

$$M\frac{\partial}{\partial M}G^{(2)}(p) = \left(-p\frac{\partial}{\partial p}-2\right)G^{(2)}(p). \tag{2.493}$$

Putting this results into the Callan-Symanzik equation for 2-point function, we then have:

$$\left(p\frac{\partial}{\partial p}-\beta\frac{\partial}{\partial\lambda}+2-2\gamma(\lambda)\right)G^{(2)}(p^2,\lambda,M) = 0. \tag{2.494}$$

This is an equation for a current. We can firstly write the solution in free theory, i.e. $\lambda = 0$ which means $\beta = \gamma = 0$:

$$G^{(2)}(p) = \frac{\mathrm{i}}{p^2}\bar{G}_0, \tag{2.495}$$

where $\bar{G}_0$ is a constant function in p^2. But one can not happy if we only have the free theory. Therefore, we need to solve the renormalization group equation with a more general form. For that, we firstly introduce some new variables to make the PDE simpler:

$$t = \ln\left(\frac{p}{M}\right), \tag{2.496}$$

$$v(x) = -\beta(\lambda), \tag{2.497}$$

$$\rho(x) = 2\gamma(\lambda) - 2, \tag{2.498}$$

$$x = \lambda. \tag{2.499}$$

Now we call the new function as $D(t, x)$, which is nothing but the $G^{(2)}$. So (2.494) becomes

$$\left(\frac{\partial}{\partial t} + v(x)\frac{\partial}{\partial x} - \rho(x)\right) D(t, x) = 0. \tag{2.500}$$

The solution of $D(t, x)$ is somehow like a current with the current density

$$j(t, x) = \rho(t, x)v(t, x). \tag{2.501}$$

So the solution of a current can be expressed as

$$D(t, x) = D_0(\bar{x}(t, x)) \exp\left[\int_0^t \mathrm{d}t' \, \rho(\bar{x}(t', x))\right], \tag{2.502}$$

with the boundary condition and the initial condition:

$$\frac{\partial}{\partial t'}\bar{x}(t', x) = -v(\bar{x}), \qquad \bar{x}(0, x) = x, \tag{2.503}$$

where $\bar{x}(t, x)$ is the position of current volum element at t and x. So the Callan-Symanzik equation has the solution:

$$G^{(2)}(p^2, \lambda, M) = \bar{G}_0(\bar{\lambda}\,(p, \lambda)) \exp\left[\int_{p'=M}^{p'=p} \mathrm{d}\ln\left(\frac{p'}{M}\right) 2\left[1 - \bar{\gamma}\left(\bar{\lambda}(p', \lambda)\right)\right]\right]. \tag{2.504}$$

It is important to note that $\bar{G}_0(\bar{\lambda}\,(p, \lambda))$ is an unknown function, which does not directly corresponds to the propagator. To fix this function, we firstly put (2.490) into (2.494):

$$0 = \left(p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda)\right)\left[\frac{\mathrm{i}}{p^2} f\left(\frac{p^2}{M^2}, \lambda\right)\right] \tag{2.505}$$

$$= \frac{\mathrm{i}}{p^2}\left(-2 + p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda)\right) f\left(\frac{p^2}{M^2}, \lambda\right) \tag{2.506}$$

Thus, we have

$$\left(p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial \lambda} - 2\gamma(\lambda)\right) f\left(\frac{p^2}{M^2}, \lambda\right) = 0. \tag{2.507}$$

So the solution is similar with $G^{(2)}$:

$$f\left(\frac{p^2}{M^2}, \lambda\right) = \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp\left[-2\int_M^p \mathrm{d}\ln\left(\frac{p'}{M}\right)\gamma(\bar{\lambda})\right]. \tag{2.508}$$

Therefore, we can take the another form of the solution by combine the solution of f and (2.490)

$$G^{(2)}(p^2, \lambda, M) = \frac{\mathrm{i}}{p^2}\bar{G}_0(\bar{\lambda}(p, \lambda)) \exp\left[-2\int_M^p \mathrm{d}\ln\left(\frac{p'}{M}\right)\gamma(\bar{\lambda}(p', \lambda))\right]. \tag{2.509}$$

Now, we can take the boundary $p = M$, which all the integration part will vanish. So we have

$$\bar{G}_0(\lambda) = -\mathrm{i}M^2 \cdot G^{(2)}(M^2, \lambda, M). \tag{2.510}$$

We can then go to the specific case and use the perturbative expansion of propagator to get each order of $\bar{G}_0(\lambda)$.

6.2 Renormalization group equation

Now the boundary condition and the initial condition in (2.503) are changed to the so-called **renormalization group equation**:

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}), \qquad \bar{\lambda}(M, \lambda) = \lambda. \tag{2.511}$$

This equation shows that the flow of the running coupling constant $\bar{\lambda}(p)$ with the speed $\beta(\bar{\lambda})$. And we set the initial energy scale is M, and then set the value at this point is the original coupling constant λ. Now we expand $\bar{\lambda}$ to the first order and use (2.471):

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}) = \frac{3}{16\pi^2}\bar{\lambda}^2 + O(\bar{\lambda}^3). \tag{2.512}$$

So we actually are solving the ODE

$$\frac{\mathrm{d}\bar{\lambda}}{\mathrm{d}t} = A\bar{\lambda}^2 \tag{2.513}$$

It is easy to solve it by integrating

$$\int_{\bar{\lambda}(M)=\lambda}^{\bar{\lambda}(p)} \bar{\lambda}'^{-2}\,\mathrm{d}\bar{\lambda}' = \int_0^{\ln(p/M)} A\,\mathrm{d}t' \tag{2.514}$$

Thus, we have

$$\frac{1}{\bar{\lambda}(p)} = \frac{1}{\bar{\lambda}} - A\ln(p/M) \tag{2.515}$$

$$= \frac{1 - A\bar{\lambda}(M)\ln(p/M)}{\bar{\lambda}(M)}. \tag{2.516}$$

Put all the parameters back, we have

$$\ln\left(\frac{p}{M}\right) = \left(\frac{3}{16\pi^2}\right)^{-1}\left(\frac{1}{\bar{\lambda}(M)} - \frac{1}{\bar{\lambda}(p)}\right). \tag{2.517}$$

Take $\bar{\lambda}(M) = \lambda$, which leads to the expression for the running coupling $\bar{\lambda}$:

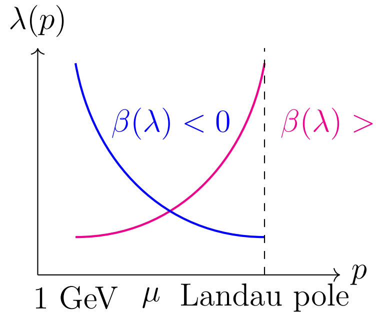
$$\bar{\lambda}(p) = \frac{\lambda}{1 - \frac{3}{16\pi^2}\lambda\ln(p/M)}. \tag{2.518}$$

So $\bar{\lambda}(p)$ grows for increasing scales p. And there is a pole when

$$1 - \frac{3}{16\pi^2}\lambda \ln\left(\frac{p}{M}\right) = 0, \tag{2.519}$$

which is called the **Landau pole**:

$$p = M \exp\left(\frac{16\pi^2}{3\lambda}\right). \tag{2.520}$$



For $p^2 \ll M^2$, theory becomes trivial:

$$\bar{\lambda}(p) \sim \frac{\bar{\lambda}(M)}{\lim\limits_{x\to 0}(1 - \ln x)} \to 0_+ \tag{2.521}$$

which corresponds to the free theory.

Generally, we have different cases for the β function, and that corresponds to the different background theory

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p,\lambda) = \beta(\bar{\lambda}) \begin{cases} > 0 & \text{IR free } (\phi^4\text{-theory, QED, ...}) \\ = 0 & \text{conformal theory } (N = 4 \text{ SYM}) \\ < 0 & \text{UV free/asymptotically free (QCD, SU(N) gauge theory).} \end{cases} \tag{2.522}$$

6.3 Higgs mass and fate of universe

As the second example, we consider the Higgs mechanism. The story begin with the Lagrangian:

$$\mathcal{L}_{\text{unbroken}} = \bar{\psi}(\mathrm{i}\gamma^\mu\partial_\mu)\psi + \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - V(\phi) + y_\psi\phi\bar{\psi}\psi. \tag{2.523}$$

where $y_\psi\phi\bar{\psi}\psi$ is the Higgs-Yukawa interaction, and we dropped all the gauge terms. To get the spontaneous symmetry breaking, we take the potential as

$$V(\phi) = \mu^2\phi^2 + \lambda\phi^4. \tag{2.524}$$

The spontaneous symmetry breaking happens when

$$\phi = H + v. \tag{2.525}$$

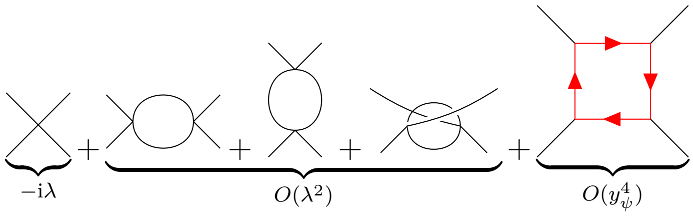
where H is the Higgs field, v is the vacuum expectation value, which in experiment measure are

$$v \simeq \frac{1}{\sqrt{\sqrt{2}G_F}} \simeq 246 \text{ GeV} \tag{2.526}$$

So plug (2.525) into the Lagrangian, we have the spontaneous symmetry breaking Lagrangian:

$$\mathcal{L}_{\text{broken}} = \bar{\psi}(\mathrm{i}\partial\!\!\!/)\psi + \frac{1}{2}(\partial_\mu H)(\partial^\mu H) - \underbrace{\lambda v^2}_{\frac{1}{2}M_H^2} H^2 - \lambda v H^3 - \frac{\lambda}{4}H^4 - \underbrace{\frac{1}{\sqrt{2}}y_\psi v}_{m_\psi}(1 + H/v)\bar{\psi}\psi \tag{2.527}$$

This mechanism turns out to the particle mass by the Higgs interact with itself and the other fields. So the coupling constant renormalization is very similar with ϕ^4 theory and Yukawa theory:



(2.528)

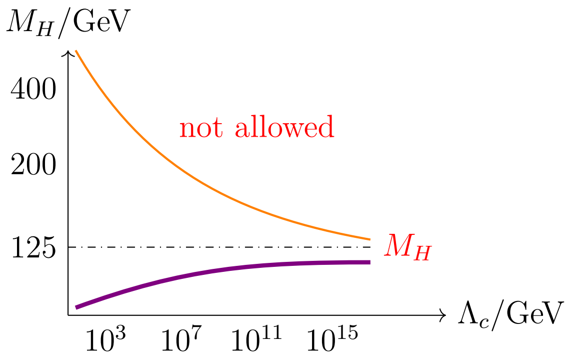
So the solution for $\lambda(Q)$ at scale $Q^2 \gg v$ is:

$$\lambda(Q) = \lambda(v)\frac{1}{1 - \frac{3}{16\pi^2}\lambda(v)\ln(Q/v)} \tag{2.529}$$

Cut off Λ_c corresponds to the Landau pole, up to which perturbative Higgs sector holds:

$$\Lambda_c = v\exp\left(\frac{16\pi^2}{3\lambda}\right) \simeq v\exp\left(\frac{16\pi^2 v^2}{M_H^2}\right) \tag{2.530}$$

where $M_H^2 = 2\lambda v^2$. So the area above the orange line are forbidden.



Now we consider fermion as the top quark, which is the heaviest fermion the standard model. It has contributions to $\beta(\lambda)$ from Higgs-Yukawa interaction, which the box diagram gives

$$Q\frac{\mathrm{d}}{\mathrm{d}Q}\lambda = \frac{1}{64\pi^2}(-3y_t^4 + 6\lambda y_t^2 + 12\lambda^2) + \dots. \tag{2.531}$$

It is interesting that the first term is negative, which means it evolves in the opposite way with the Higgs. Solving this equation for small λ ($\lambda \ll 1$), we have

$$\lambda(Q) = \lambda(v) + \frac{1}{32\pi^2}\left(-\frac{3m_t^4}{v^4}\right)\ln\left(\frac{Q}{v}\right) \tag{2.532}$$

since $m_t = y_t\frac{v}{\sqrt{2}}$, $m_t \sim 172$ GeV, so in principle, $\lambda(Q)$ can be negative at large Q! That means the vacuum will be super unstable. So to avoid that, we require a bound on Higgs mass:

$$M_H^2 \geq \frac{v^2}{16\pi^2}\left(-\frac{3m_t^4}{v^4}\right)\ln\left(\frac{Q}{v}\right), \tag{2.533}$$

such that we can keeping $\lambda(Q) > 0$.

7. Renormalization for local operators and evolution of mass parameter

We first introduce the idea of local operator. The local operator is multiple field operator compositing in the same space-time position, namely the production of fields. This will happen when the energy scale is much lower than the high energy scattering, e.g. weak decay.So to understand this, we can recall the beta decay in electroweak interaction.

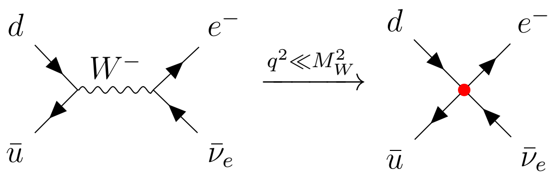
The electroweak interaction between left fermions and W-boson is:

$$\delta\mathcal{L} = \frac{g_{\text{ew}}}{\sqrt{2}} W_\mu \bar{\psi}\gamma^\mu(1-\gamma_5)\psi. \tag{2.534}$$

And the propagator of W-boson is

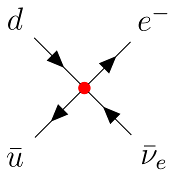
$$\mu \overset{W^-}{\sim\!\sim\!\sim\!\sim\!\sim} \nu \;=\; \frac{-\mathrm{i}g^{\mu\nu}}{q^2 - M_W^2 + \mathrm{i}\epsilon} \tag{2.535}$$

In the low energy limit $q^2 \ll M_W^2$, the vertex and propagator of β-decay reduce to the 4-Fermi interaction vertex:



. (2.536)

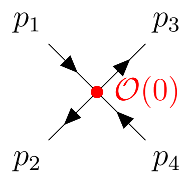
So we can reduce a process propagate from μ to ν to the process happens in a fixed point due to the reaction happening so fast, i.e.:



$$= \frac{g_{\text{ew}}^2}{2M_W^2}(\bar{\psi}\gamma^\mu(1-\gamma_5)\psi)_{\text{quark}}(\bar{\psi}\gamma_\mu(1-\gamma_5)\psi)_{\text{lep}} = \frac{g_{\text{ew}}^2}{2M_W^2}\mathcal{O}(x). \tag{2.537}$$

Generally, this special vertex can be seen as a Green's function:

$$\langle\psi(p_1)\bar{\psi}(p_2)\psi(p_3)\bar{\psi}(p_4)\mathcal{O}(0)\rangle = \tag{2.538}$$



7.1 Renormalized Green's functions with local operator $\mathcal{O}$ in ϕ^4-theory

Now we consider the renormalization of the local operator. Since the vertex is described by the Green's function, we can then consider a general form:

$$\begin{aligned} G^{(n,\mathcal{O})}(p_1 \dots p_n, k) &= \langle\phi(p_1)\dots\phi(p_n)\mathcal{O}(k)\rangle && (2.539)\\ &= Z^{-n/2}(\mu)\underbrace{Z_{\mathcal{O}}^{-1}(M)}_{\substack{\text{operator}\\ \text{renormalization}}}\langle\underbrace{\phi_0(p_1)\dots\phi_0(p_n)}_{\text{bare fields}}\underbrace{\mathcal{O}_0(k)}_{\substack{\text{bare}\\ \text{operator}}}\rangle && (2.540)\\ &= Z^{-n/2}(\mu)Z_{\mathcal{O}}^{-1}(M)G_0^{(n,\mathcal{O})}(p_1 \dots p_n, k) && (2.541)\end{aligned}$$

where $\mathcal{O} = \phi^2$ or ϕ^4 or ... Tthe lower index "0" represent the bare quantity, and M is the renormalization scale. The bare operator and the renormalized operator have the relation:

$$\mathcal{O}_{\text{bare}} = Z_{\mathcal{O}}(M)\mathcal{O}_{\text{ren}}. \tag{2.542}$$

So we can derive the Callan-Symanzik equation by using

$$\begin{aligned} 0 &= M\frac{\mathrm{d}}{\mathrm{d}M}G_0^{(n,\mathcal{O}_0)} && (2.543)\\ &= M\frac{\mathrm{d}}{\mathrm{d}M}\left(Z^{n/2}Z_{\mathcal{O}}G^{(n,\mathcal{O})}\right) && (2.544)\\ &= \left(M\frac{\partial}{\partial M}Z^{n/2}\right)Z_{\mathcal{O}}G^{(n,\mathcal{O})} + Z^{n/2}\left(M\frac{\partial}{\partial M}Z_{\mathcal{O}}\right)G^{(n,\mathcal{O})} + Z^{n/2}Z_{\mathcal{O}}\left(M\frac{\partial}{\partial M}G^{(n,\mathcal{O})}\right), && (2.545)\end{aligned}$$

dividing $Z^{n/2}Z_{\mathcal{O}}$ on both side, we have

$$\begin{aligned} 0 &= \left(\frac{M}{Z^{n/2}}\frac{\partial Z^{n/2}}{\partial M}\right)G^{(n,\mathcal{O})} + \left(\frac{M}{Z_{\mathcal{O}}}\frac{\partial Z_{\mathcal{O}}}{\partial M}\right)G^{(n,\mathcal{O})} + M\frac{\partial}{\partial M}G^{(n,\mathcal{O})} && (2.546)\\ &= \left(M\frac{\partial}{\partial M}\ln Z^{n/2}\right)G^{(n,\mathcal{O})} + \left(M\frac{\partial}{\partial M}\ln Z_{\mathcal{O}}\right)G^{(n,\mathcal{O})} + M\frac{\partial}{\partial M}G^{(n,\mathcal{O})}, && (2.547)\end{aligned}$$

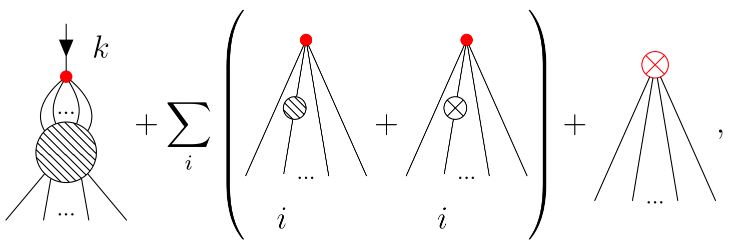
where we can define the anomalous dimension respectively for field and local operator as:

$$\begin{aligned} \gamma &= \frac{M}{2}\frac{\partial}{\partial M}\ln Z_{\lambda}(M) && (2.548)\\ \gamma_{\mathcal{O}} &= M\frac{\partial}{\partial M}\ln Z_{\mathcal{O}}(M). && (2.549)\end{aligned}$$

Then we have the Callan-Symanzik equation:

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial\lambda} + n\gamma(\lambda) + \gamma_{\mathcal{O}}(\lambda)\right)G^{(n,\mathcal{O})} = 0, \tag{2.550}$$

Now we compute $\gamma_{\mathcal{O}}$ for N-point Green's function. The most general case can be expressed as

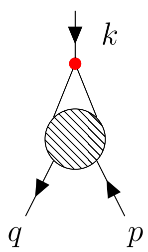


(2.551)

where the red point and red cross represent the operator vertex and the operator counterterm.

7.2 Renormalization for local operators in massless ϕ^4-theory

As an example to help us understanding this. We consider $\mathcal{O} = \phi^2$ inserted in two-point function:



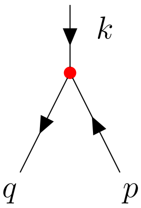
$$= \langle\Omega|\,\phi(q)\phi(p)\phi^2(k)\,|\Omega\rangle = G^{(2,\phi^2)}. \tag{2.552}$$

Thus, we can expand it as the normal Dyson series:

$$G^{(2,\phi^2)}(q,p;k) = \sum_{m=0}^{\infty} \frac{\mathrm{i}^m}{m!} \int \left[\prod_{a=1}^{m} \mathrm{d}^d x_a\right] \left(-\frac{\lambda}{4!}\right)^m \langle 0|T\{\phi(q)\phi(p)\phi^2(k)[\phi^4(x_1)\ldots\phi^4(x_m)]\}|0\rangle_{\text{connect}}. \tag{2.553}$$

The leading order, namely the tree-level non-amputated Green's function is just simply the contraction of the field and the operator:

$$G^{(2,\phi^2)}_{\text{tree}}(q,p;k) =$$



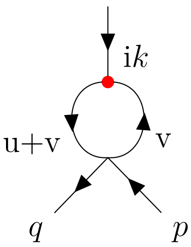
$$= \int \mathrm{d}^d x\, \mathrm{e}^{-\mathrm{i}k\cdot x}\langle 0|T\{\phi(q)\phi(p)\phi_A(x)\phi_B(x)\}|0\rangle \tag{2.554}$$

$$= \int \mathrm{d}^d x\, \mathrm{e}^{-\mathrm{i}k\cdot x}\left[\phi(q)\phi(p)\phi_A(x)\phi_B(x) + \phi(q)\phi(p)\phi_A(x)\phi_B(x)\right] \tag{2.555}$$

$$= \left(\frac{\mathrm{i}}{q^2}\right)\left(\frac{\mathrm{i}}{p^2}\right)\cdot 2. \tag{2.556}$$

To consider the renormalization for the operator, we need to compute 1-PI contribution at 1-loop (non-amputated):

$$G^{(2,\phi^2)}_{\text{1-loop}}(q,p;k) =$$



$$= \int \mathrm{d}^d x \mathrm{d}^d z\, \mathrm{e}^{-\mathrm{i}k\cdot x}\left\langle 0\left|T\left\{\phi(q)\phi(p)\left[\frac{-\mathrm{i}\lambda}{4!}\phi^4(z)\right]\left[\frac{1}{2}\phi_A(x)\phi_B(x)\right]\right\}\right|0\right\rangle \tag{2.557}$$

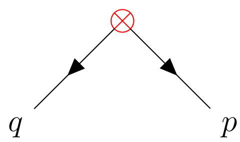
$$= \frac{-\mathrm{i}\lambda}{2}\int \mathrm{d}^d x \mathrm{d}^d z\, \mathrm{e}^{-\mathrm{i}k\cdot x}\left[\phi(q)\,\phi(p)\quad \phi(z)\,\phi(z)\,\phi(z)\,\phi(z)\quad \phi_A(x)\,\phi_B(x)\right] \tag{2.558}$$

$$= \left(\frac{\mathrm{i}}{q^2}\right)\left(\frac{\mathrm{i}}{p^2}\right)\int \frac{\mathrm{d}^d v}{(2\pi)^d}(-\mathrm{i}\lambda)\frac{\mathrm{i}}{v^2}\frac{\mathrm{i}}{(k+v)^2}, \tag{2.559}$$

with some calculation, one has:

$$G^{(2,\phi^2)}_{\text{1-loop}}(q,p;k) = \frac{\mathrm{i}}{p^2}\frac{\mathrm{i}}{q^2}\left(-\frac{\lambda}{(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})}{\Delta^{2-d/2}}\right) \tag{2.560}$$

with the counterterm:



$$= \frac{\mathrm{i}}{p^2}\frac{\mathrm{i}}{q^2}2\delta\phi^2 \tag{2.561}$$

Comparing of coefficients in these two graphs, we then can find the counterterm at scale $p^2 = -M^2$ should be:

$$\delta\phi^2 = \frac{\lambda}{2(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})}{(M^2)^{2-d/2}}. \tag{2.562}$$

This implies the operator renormalization is

$$Z_{\phi^2} = 1 - \delta\phi^2 = 1 - \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2-\frac{d}{2})}{(M^2)^{2-d/2}} \tag{2.563}$$

$$\overset{d\,=\,4\,-\,\varepsilon}{=} 1 - \frac{\lambda}{(4\pi)^2}\frac{1}{\epsilon} + \text{finite terms.} \tag{2.564}$$

Using $\ln(1+x) \approx x$, we can obtain the anomalous dimension of the operator in $d = 4 - \varepsilon$:

$$\gamma_{\phi^2} = M\frac{\partial}{\partial M}\ln Z_{\phi^2} = \frac{-\lambda}{2(4\pi)^{d/2}}\Gamma\left(2 - \frac{d}{2}\right) \cdot \left[-(4-d)M^{-(4-d)}\right] \tag{2.565}$$

$$\overset{d\,=\,4\,-\,\varepsilon}{=} \frac{-\lambda}{2(4\pi)^{2-\epsilon/2}}\Gamma\left(\frac{\epsilon}{2}\right) \cdot \left[-\epsilon M^{-\epsilon}\right] \tag{2.566}$$

$$= \frac{\lambda}{16\pi^2}. \tag{2.567}$$

More generally, if we insert the operator into an n-point Green's function, we then need to consider the field renormalization with the operator renormalization $Z_\phi^{n/2} Z_{\mathcal{O}}$, therefore the anomalous dimension becomes:

$$\gamma_{\mathcal{O}}(\lambda) = M\frac{\partial}{\partial M}\left(-\delta_{\mathcal{O}} + \frac{n}{2}\delta_Z\right), \tag{2.568}$$

with the approximation:

$$\ln\left(Z_\phi^{n/2} Z_{\mathcal{O}}\right) = \frac{n}{2}\ln Z_\phi + \ln Z_{\mathcal{O}} \approx \frac{n}{2}\delta_Z - \delta_{\mathcal{O}}. \tag{2.569}$$

This case can be treated as renormalized ϕ^4-theory with effective operator $-\frac{1}{2}m^2\phi^2$, where ϕ^2 is local operator inserted in Green's function. So the Callan-Symanzik equation with mass anomalous dimension $m(\mu)$ is

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\phi^2} m^2 \frac{\partial}{\partial m^2}\right) G^{(n,\phi^2)}\left(\{p_i\}, M, \lambda, m\right) = 0 \tag{2.570}$$

7.3 Evolution of mass parameter

Moreover, one can write the generalization to local operators $\mathcal{O}_i$ in ϕ^4-theory as:

$$\mathcal{L}(C_i) = \mathcal{L}_{\text{eff}} + \sum_i C_i \mathcal{O}_i(x) \tag{2.571}$$

where C_i is the coupling of local operators. So the Callan-Symanzik equation is

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \gamma_i(\lambda) C_i \frac{\partial}{\partial C_i}\right) G^{(n)}(M, \lambda, \{C_i\}) = 0 \tag{2.572}$$

To guarantee the dimension of Lagrangian is $[\text{Mass}]^4$ in 4-dimension spacetime, so we need to explicit mass dimension in coupling C_i as:

$$C_i = \rho_i M^{4-d_i} \tag{2.573}$$

where ρ_i is dimensionless, d_i is dimension of $\mathcal{O}_i$. Therefore, one can see that the variables change to $(M, \lambda, \{\rho_i\})$. To make the partial derivative independent to ρ_i, we have

$$M\frac{\partial}{\partial M}\bigg|_{\rho_i} = M\frac{\partial}{\partial M}\bigg|_{C_i} + \sum_i \left(M \frac{\partial C_i}{\partial M}\bigg|_{\rho_i}\right)\frac{\partial}{\partial C_i} \tag{2.574}$$

Notice that

$$\rho_i \frac{\partial}{\partial \rho_i} = \rho_i M^{4-d_i} \frac{\partial}{\partial C_i} = C_i \frac{\partial}{\partial C_i} \tag{2.575}$$

so we have

$$M \frac{\partial}{\partial M}\Big|_{C_i} = M \frac{\partial}{\partial M}\Big|_{\rho_i} - \sum_i (4 - d_i)\rho_i \frac{\partial}{\partial \rho_i} \tag{2.576}$$

Then the Callan-Symanzik equation becomes:

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i (\gamma_i(\lambda) + d_i - 4)\rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0, \tag{2.577}$$

where we can define the β function for new coupling ρ_i:

$$\beta_i(\lambda) = (\gamma_i(\lambda) + d_i - 4)\rho_i, \tag{2.578}$$

then the Callan-Symanzik equation can be reduced as:

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \beta_i \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0. \tag{2.579}$$

With this, we can easily solve the running coupling ρ_i from the renormalization group equation for β_i

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)} \bar{\rho}_i = \beta_i(\bar{\rho}_i, \lambda). \tag{2.580}$$

Since β_i in the low energy limit, namely $\gamma_i(\lambda) \ll 1$, is dominated by the classical dimension $d_i - 4$, we then have

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)} \bar{\rho}_i \simeq (d_i - 4 + \dots).\bar{\rho}_i \tag{2.581}$$

Therefore, the solution in low energy limit reads:

$$\bar{\rho}_i = \rho \left(\frac{p}{M} \right)^{d_i - 4}. \tag{2.582}$$

Now we can clearly see that for the operators with mass dimension $d_i > 4$ will vanish as $p \to 0$, e.g., ϕ^6 theory with $d_i = 6$. These operators are called the **irrelevant operators**, which usually decrease fast until vanish in low energy limit, namely one can never feel these operators' contribution. On the contrary, the **relevant operators** are the operators with mass dimension $d_i < 4$, e.g., the mass term $\frac{1}{2}m^2\phi^2$, which will exponentially increase in low energy limit. And the operators like ϕ^4 are called the **marginal operators**, which do not changing with the energy scale due to $d_i = 4$. This type operators are only determined by the anomalous dimension $\gamma_i(\lambda)$.

This reminds us that the effective field theory truly works even though we don't know all the details in high energy level. But we still can derive the precise renormalizable Lagrangian in low energy level.